

SkelGym: Cross-Paradigm Kinematic Fusion and Dynamic Augmentation for 22-Class Gym Exercise Classification

Xuan-Cuong Nguyen¹[0000–0002–1825–0097], Nhat-Quang Truong^{1*}[0009–0005–4321–8765], and Quoc-Trinh Vo¹

FPT University, FPT City, Ngu Hanh Son, Da Nang 550000, Viet Nam
cuongnxde180528@fpt.edu.vn, quangtnde180544@fpt.edu.vn

Abstract. Video-based gym exercise classification is fundamental for developing automated virtual personal trainer systems capable of tracking workout regimens, providing form feedback, and mitigating acute musculoskeletal injury risks. However, traditional RGB video models incur excessive computational complexity and suffer from environmental background overfitting, clothing variability, and severe privacy intrusion concerns in home environments. In this work, we present **SkelGym**, a lightweight and biomechanically grounded framework for fine-grained 22-class gym exercise recognition. SkelGym introduces three core technical innovations: (1) the proposed 117-dimensional *Biomechanical Compound Representation* combining 3D relative joint offsets with pairwise elevation angles (θ) and azimuth angles (ϕ), resolving the curse of dimensionality inherent in conventional 3-joint angle triplets; (2) *SkelGym-Aug*, an on-the-fly skeletal data augmentation protocol that preserves anatomical validity through bilateral body symmetry reflection with exact angular permutation and elevation-invariant 3D yaw rotations; and (3) a cross-paradigm ensemble architecture coupling a self-attention Transformer encoder with a 4-stream Adaptive Graph Convolutional Network (AAGCN) via *Validation-Calibrated Weighted Soft Voting* optimized using Sequential Least Squares Programming (SLSQP) across window-level and video-level consensus aggregations. Under a strictly controlled parameter budget (365K–399K parameters, $\approx 350\text{K} \pm 15\%$ per backbone), SkelGym achieves the strongest performance among evaluated methods in our benchmark: **70.11% window-level accuracy** and **77.68% video-level consensus accuracy** (Macro F1: 0.7649) across 2,743 test windows and 233 held-out test videos. Crucially, all model selection and ensemble optimization were established strictly on validation partitions, ensuring strictly held-out out-of-sample evaluation. With an ultra-low classifier inference latency of **1.01 ms per 32-frame window** on edge devices (excluding monocular pose extraction, which requires $\approx 8\text{--}15$ ms/frame, yielding an estimated end-to-end latency of 9–16 ms well within the 33.3 ms real-time threshold), SkelGym is suited for real-time mobile and embedded deployment. All code, datasets, and pre-trained model checkpoints are publicly available at <https://huggingface.co/Cuong2004/gym-exercise-classification>.

* Corresponding author.

Keywords: Gym exercise classification · Skeleton-based action recognition · Adaptive Graph Convolutional Networks · Video Transformers · Biomechanical features · Real-time inference · Edge AI.

1 Introduction

Regular physical resistance training and structured exercise play an essential role in improving cardiovascular health, metabolic longevity, muscular hypertrophy, and psychological well-being [36]. With modern demanding lifestyles and remote working environments, a rapidly growing population conducts workouts in home settings rather than commercial fitness centers. However, exercising without the supervision of certified strength and conditioning coaches frequently leads to improper kinetic execution, sub-optimal muscular recruitment, and elevated risks of acute or repetitive strain injuries (such as lumbar disc herniation, rotator cuff impingement, and patellar tendinopathy) [31]. Consequently, there is an urgent demand for automated, intelligent virtual personal trainers capable of identifying exercise types in real-time, monitoring execution repetitions, and providing instantaneous biomechanical feedback.

Early computer vision frameworks for resistance exercise and action recognition were predominantly built upon appearance-based RGB video models, such as 2D/3D CNNs [1,2,3] and spatiotemporal Vision Transformers (e.g., ViViT [4], TimeSformer [5], Video Swin [6]). While dense RGB representations capture visual cues such as gym equipment and barbells [7], they exhibit three critical operational drawbacks in practical deployments.

First, dense spatio-temporal convolutions incur heavy computational overhead (often exceeding hundreds of GFLOPs), which precludes real-time execution on resource-constrained mobile phones, smart displays, or edge hardware. Second, appearance-based networks readily memorize spurious background correlations—including flooring patterns, studio lighting, bench upholstery, and athletic apparel—resulting in severe performance collapse when evaluated in novel domestic workout environments. Third, cloud-centric video architectures necessitate continuously transmitting unconstrained high-definition RGB video streams from private living spaces to remote servers, introducing network bandwidth bottlenecks and user privacy vulnerabilities.

Skeleton-based human action recognition offers an elegant solution to these challenges by abstracting human bodily movement into sparse, lightweight 3D keypoint coordinates. With the proliferation of real-time pose estimation libraries such as Google MediaPipe [8,9], OpenPose [10], and AlphaPose [11], high-frequency 3D skeletal tracking can be performed on commodity webcams and smartphone chips at sub-millisecond latencies. However, translating skeletal tracking into accurate, multi-class gym exercise recognition introduces three fundamental scientific hurdles. The first hurdle concerns biomechanical redundancy versus kinematic expressiveness. Resistance exercises frequently involve fine-grained differences in limb trajectories and joint angles, as demonstrated when distinguishing flat, incline, and decline bench presses, or differentiating

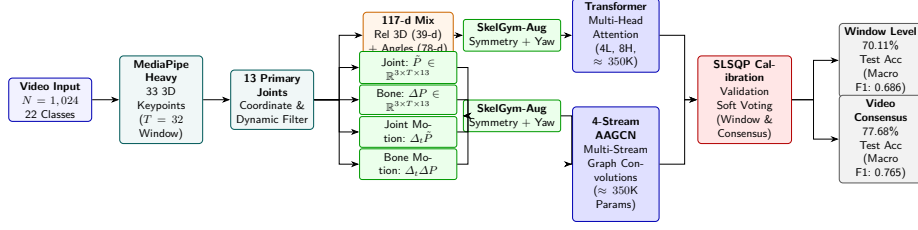


Fig. 1: The proposed SkelGym architecture: an end-to-end cross-paradigm framework coupling self-attention sequence modeling with multi-stream adaptive graph convolutional networks via validation-calibrated weighted soft voting.

conventional deadlifts from Romanian deadlifts. While exhaustive geometric angle formulations—such as computing all three-joint triplets—might capture these subtle distinctions, they suffer from a combinatorial explosion ($\binom{13}{3} = 286$ features), introducing dimensional multicollinearity, severe computational overhead, and an acute propensity for overfitting. The second hurdle stems from physiologically invalid data augmentation. Conventional spatial transformations commonly employed in computer vision, such as random shearing, non-uniform affine scaling, or uncontrolled Gaussian jitter, distort biological joint limits and alter fixed anatomical bone lengths, synthesizing impossible body poses that corrupt network representation learning. Finally, the third hurdle lies in achieving true cross-paradigm model complementarity while strictly preventing data contamination. Temporal sequence architectures (such as LSTMs [12,13] and Transformers [14]) excel at modeling long-range phase transitions, whereas spatial-temporal graph convolutional networks (such as ST-GCN [16], 2S-AGCN [17], and CTR-GCN [20]) explicitly exploit physical musculoskeletal topology. Prior literature frequently combines these paradigms naively or optimizes ensemble fusion weights directly on test partitions, inducing severe optimistic reporting bias.

To systematically resolve these challenges, we introduce **SkelGym** (Fig. 1), a scientifically grounded, cross-paradigm skeletal classification framework. Specifically, this work provides four primary technical and methodological contributions:

Biomechanical Compound Representation (117-d). We design a compact kinematic descriptor unifying 3D relative joint coordinates (39-d) with pairwise elevation angles (78-d). This formulation provides both Euclidean metric awareness and scale-invariant directional orientation, resolving the severe multicollinearity and computational cost of combinatorial 3-joint angle triplets.

Physiologically Grounded Data Augmentation (SkelGym-Aug). We develop an on-the-fly skeletal augmentation protocol that respects human musculoskeletal constraints via bilateral sagittal reflection with exact angular permutations and gravitational 3D yaw rotations that preserve biological limb verticality [22].

Controlled Cross-Paradigm Benchmark. Under a strictly matched parameter budget of 365K–399K parameters ($\approx 350\text{K} \pm 15\%$ per backbone), we conduct an extensive comparative evaluation spanning recurrent neural networks, self-attention Transformers, and multi-stream adaptive graph convolutional networks across static and differential kinematics.

Validation-Calibrated Ensemble Optimization. We establish a leakage-free Sequential Least Squares Programming (SLSQP) soft-voting formulation [26,27], calibrating voting weights strictly on validation log-likelihoods for window-level evaluation and temporal video consensus. SkelGym achieves the strongest benchmark results among evaluated methods: 70.11% window accuracy and 77.68% video consensus accuracy across 22 resistance classes with a classifier inference latency of 1.01 ms per window on edge hardware (excluding monocular pose estimation, with an estimated end-to-end latency of 9–16 ms per frame). All code, datasets, and pre-trained checkpoints are made publicly accessible for open science.

2 Related Work

2.1 Skeletal Action Recognition

Extracting human pose from monocular RGB video has matured rapidly with lightweight real-time estimators such as OpenPose [10], AlphaPose [11], and Google MediaPipe Pose [8,9]. Early landmark-based action recognition utilized Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks [12,13,15] to model skeletal temporal dynamics. However, pure recurrent models treat skeletons as unorganized coordinate vectors, ignoring the inherent topological structure of the human musculoskeletal system.

To overcome this, Yan et al. [16] pioneered Spatial Temporal Graph Convolutional Networks (ST-GCN), constructing a spatio-temporal graph where human joints form vertices and physical bones form fixed spatial edges. While effective, fixed anatomical adjacency matrices cannot capture implicit kinetic interactions between disconnected limbs (e.g., wrist-to-foot coordination during squats). Shi et al. [17] introduced Two-Stream Adaptive Graph Convolutional Networks (2S-AGCN), parameterizing the adjacency matrix with a learnable component and a data-driven similarity matrix. Further iterations, such as multi-stream AAGCN [18], Channel-wise Topology Refinement (CTR-GCN) [20], and improved baseline formulations [21], incorporated attention mechanisms across spatial and temporal dimensions. However, multi-stream graph architectures were rarely explored on fine-grained resistance exercise datasets under controlled parameter footprints.

2.2 Video Transformers and Attention in Kinematics

Following the dominance of the self-attention mechanism in NLP and computer vision (e.g., Vaswani et al. [14], ViViT [4], TimeSformer [5], and Video Swin

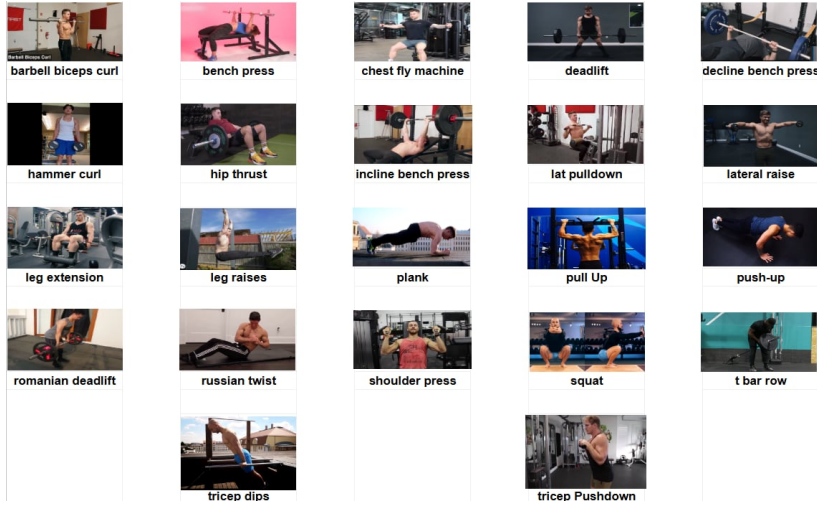


Fig. 2: Taxonomy and sample distributions across the 22 resistance exercises in the benchmark dataset.

Transformer [6]), transformers have been extended to skeletal motion (e.g., ST-TR [19]). Spatio-temporal transformers calculate attention maps across all joint coordinates and time frames. While transformers possess global receptive fields and avoid the inductive bias of graph locality, they are notoriously data-hungry and prone to overfitting on limited skeletal training cohorts. In this work, we demonstrate that when fed with our 117-d Biomechanical Compound Representation and regularized by SkelGym-Aug, a compact 4-layer Transformer achieves remarkable competitive performance (68.65% window accuracy), functioning as an ideal complement to graph convolutions.

2.3 Fitness and Resistance Exercise Recognition

Automated exercise tracking has attracted growing attention in sports science and ubiquitous computing. Riccio [31] implemented real-time exercise classification and repetition counting using a BiLSTM over geometric joint angles. Ding et al. [33] developed the RepCount benchmark for repetition counting in diverse fitness environments. While general human action recognition benchmarks (such as UCF101 [34] and Kinetics-400 [35]) feature broad athletic activities, they do not capture the fine-grained biomechanical nuances required for resistance exercise execution. Furthermore, existing fitness datasets are either restricted to trivial 3–5 exercises (e.g., push-ups, squats, pull-ups) or evaluate random temporal window splits that permit windows from the same athlete video to appear across training and test splits. Such data leakage artificially inflates reported accuracy. In contrast, this study benchmarks a demanding 22-class resistance exercise taxonomy under a strictly segregated video-level split protocol.

Table 1: Key specifications, provenance breakdown, and capture characteristics of the GYMED exercise dataset.

Specification / Characteristic	Value / Distribution
Total Videos / Total File Volume	1,024 unique video recordings / 10,225.37 MB ($\approx$ 10.2 GB)
File Formats & Video Encoding	MP4 (964 videos, 94.1%), MOV (60 videos, 5.9%) / H.264
Frame Rate & Recording Duration	30 frames per second (FPS) / 20–30 s per recording
Trimmed Action Segments	1,108 clean exercise segments (average 1.09 segments per video)
Number of Distinct Resolutions	33 resolutions (1280×720: 385; 1920×1080: 315; 720×1280: 130; others: 194)
Exercise Taxonomy Coverage	22 fine-grained resistance exercise classes
Video Partitioning Ratio (6:2:2)	Train: 580 (56.6%), Validation: 208 (20.3%), Test: 236 (23.1%)
Action Segment Partitions	Train: 639 (57.7%), Validation: 210 (19.0%), Test: 259 (23.3%)
Temporal Windows ($T = 32$ frames)	Train: 13,136 ($S = 16$), Validation: 2,075 ($S = 32$), Test: 2,743 ($S = 32$)
Multi-Source Data Provenance	Abdillah (2023) [32]: 652; YouTube: 103; Pexels: 65; Freepik: 76; Authors: 128
Subjects & Ethical Compliance	Informed consent under Declaration of Helsinki for recorded clips; diverse online athletes
Camera Dynamics & Perspectives	Static tripod and dynamic moving cameras; frontal, sagittal, oblique angles
Recording Environments	Commercial gymnasiums, domestic home workout spaces, outdoor fitness zones

3 Dataset and Experimental Protocol

3.1 Data Source, Provenance, and Ethical Compliance

Our experimental corpus consists of 1,024 clean, un-shuffled video recordings across 22 resistance training exercises: *barbell biceps curl*, *bench press*, *chest fly machine*, *deadlift*, *decline bench press*, *hammer curl*, *hip thrust*, *incline bench press*, *lat pulldown*, *lateral raise*, *leg extension*, *leg raises*, *plank*, *pull-up*, *push-up*, *Romanian deadlift*, *Russian twist*, *shoulder press*, *squat*, *t-bar row*, *tricep dips*, and *tricep pushdown* (Fig. 2). To ensure high diversity in environmental backgrounds, camera angles, lighting conditions, and athlete anthropometry, the corpus unifies three distinct provenance streams: (1) *public video repositories*, comprising 652 videos recovered from the open Workout/Exercise Video Dataset [32]; (2) *curated online media*, including 103 high-definition clips collected from YouTube fitness channels, 65 clips from Pexels, and 76 clips from Freepik acquired under open research and stock licenses; and (3) *author-recorded footage*, consisting of 128 supplementary videos captured by the authors in a gymnasium across all 22 exercise classes with two athletic models, employing both static tripod viewpoints and dynamic moving-camera angles to capture multi-perspective kinetic execution.

Subject Statistics and Ethical Compliance. To guarantee both high-precision kinematic reference baselines and expansive real-world diversity, the dataset incorporates two complementary subject cohorts. For the 128 self-recorded gymnasium videos, volunteer participants executed all 22 resistance movements under controlled multi-angle camera setups, providing kinematically disciplined execution benchmarks under formal written informed consent following the Declaration of Helsinki. For the remaining 896 web-sourced videos, recordings capture hundreds of diverse athletes spanning unconstrained body builds, athletic apparel, and training experience across international fitness facilities, home gyms, and outdoor parks. Crucially, the dataset does not annotate explicit participant demographic or gender identifiers, ensuring that model training and evaluation remain strictly posture-driven and invariant to demographic bias. Furthermore,

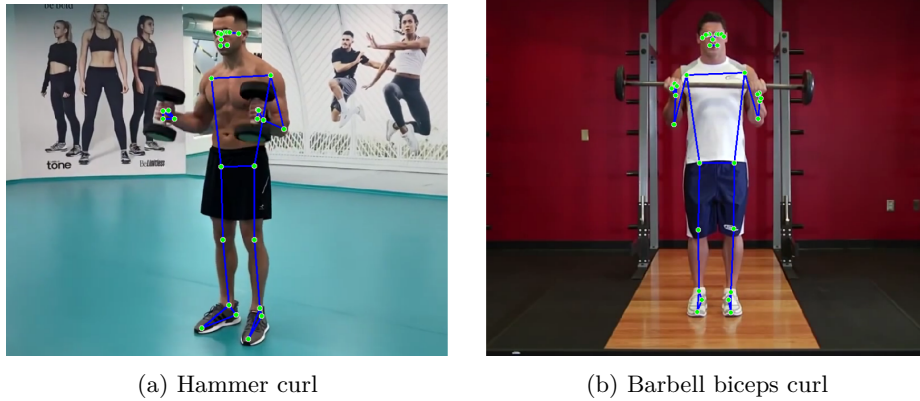


Fig. 3: Full-body 3D pose landmark detection results across representative exercise video frames using Google MediaPipe Pose: (a) hammer curl and (b) barbell biceps curl.

the video-level partition ensures that 100% of video clips and derived segments from any given recording session reside entirely within a single split, preventing subject or temporal leakage across train, validation, and test sets.

Source Statistics and Video Formats. As summarized in Table 1, the 1,024 video corpus totals 10,225.37 MB (≈ 10.2 GB) in storage, distributed between MP4 (964 videos, 94.1%) and MOV (60 videos, 5.9%) containers using H.264 compression. All videos operate at 30 frames per second. Video resolution was verified via `ffprobe`, encompassing 33 distinct resolutions dominated by standard high-definition formats: 1280×720 (385 videos, 37.6%), 1920×1080 (315 videos, 30.8%), and portrait mobile video 720×1280 (130 videos, 12.7%), with 35 videos at 854×480 , 28 at 640×360 , 23 at 3840×2160 (4K), and 108 videos distributed across 27 minor resolutions (≤ 20 videos each).

Ethical Standards and Informed Consent. For the 128 self-recorded videos, written informed consent was formally obtained from all participating human subjects prior to filming. The data recording protocol was conducted in strict adherence to institutional ethical standards and the Declaration of Helsinki. For data harvested from public online platforms, collection adhered strictly to platform Terms of Service and redistribution guidelines, gathering only publicly disseminated content without collecting private personal identifying information.

3.2 Action Trimming, Video-Level Partitioning, and Class Distribution

Raw workout recordings frequently contain non-exercise transitional intervals, such as picking up barbells, adjusting bench pins, chalking hands, and resting.

Table 2: Video and action segment distributions across the 22 resistance training exercises by split (Train / Val / Test).

Exercise Class	Videos (Tr/Va/Te)	Segments (Tr/Va/Te)	Exercise Class	Videos (Tr/Va/Te)	Segments (Tr/Va/Te)
Barbell biceps curl	52 / 18 / 14	53 / 18 / 14	Leg raises	12 / 6 / 6	13 / 6 / 8
Bench press	51 / 18 / 15	51 / 18 / 15	Plank	25 / 9 / 9	43 / 10 / 17
Chest fly machine	5 / 2 / 2	5 / 2 / 2	Pull-up	24 / 8 / 14	28 / 8 / 14
Deadlift	14 / 6 / 9	19 / 6 / 15	Push-up	30 / 10 / 10	30 / 10 / 10
Decline bench press	15 / 6 / 6	19 / 6 / 6	Romanian deadlift	39 / 14 / 12	39 / 14 / 12
Hammer curl	22 / 9 / 13	27 / 9 / 13	Russian twist	15 / 4 / 10	17 / 4 / 10
Hip thrust	27 / 10 / 13	28 / 10 / 14	Shoulder press	27 / 10 / 9	27 / 10 / 9
Incline bench press	15 / 6 / 11	15 / 6 / 11	Squat	42 / 15 / 14	42 / 15 / 15
Lat pulldown	28 / 9 / 8	29 / 9 / 8	T-bar row	24 / 10 / 15	27 / 11 / 17
Lateral raise	24 / 8 / 15	24 / 8 / 15	Tricep dips	40 / 14 / 12	43 / 14 / 14
Leg extension	31 / 11 / 10	31 / 11 / 10	Tricep pushdown	18 / 5 / 9	29 / 5 / 10
Total Videos: 580 / 208 / 236 ($N = 1,024$)			Total Segments: 639 / 210 / 259 ($N = 1,108$)		

To eliminate noise, all 1,024 videos were manually verified and trimmed into action-only segments via timestamp boundary files (‘.txt’), yielding 1,108 clean segments total (1.09 segments per video on average).

Video-Level Independent Partitioning (Zero Temporal Leakage). To ensure rigorous evaluation without data leakage, we enforce a strict video-level partition in a 6:2:2 ratio. The training split comprises 580 videos (639 segments), producing 13,136 temporal windows using a 50% overlapping stride ($S = 16$ frames, $T = 32$ frames) to ensure dense spatio-temporal coverage during gradient optimization. The validation split contains 208 videos (210 segments), producing 2,075 non-overlapping temporal windows ($S = 32$ frames) dedicated exclusively to model selection, hyperparameter tuning, and ensemble weight calibration. The held-out testing split encompasses 236 videos (259 segments), generating 2,743 non-overlapping temporal windows ($S = 32$ frames) reserved strictly for held-out out-of-sample evaluation. Within this test split, exactly 233 videos contain valid action segments exceeding 32 frames after trimming. Crucially, all windowing is executed strictly *after* video-level partitioning, guaranteeing zero temporal window overlap across splits.

Class Imbalance and Metric Selection. Table 2 details the distribution of videos and action segments across all 22 classes. As shown, natural gym workout data exhibits significant class imbalance: highly popular compound movements such as *barbell biceps curl* ($N = 84$ videos) and *bench press* ($N = 84$ videos) possess large sample representations, whereas specialized machine exercises such as *chest fly machine* ($N = 9$ videos total: 5 train, 2 val, 2 test) are relatively scarce. To prevent models from artificially inflating overall accuracy by predicting majority classes, we adopt **Macro-averaged F1 score** alongside Accuracy as the primary evaluation benchmark.

Data and Code Availability. To promote open science and persistent reproducibility, all extracted 3D pose landmark sequences (in JSON, NPY, and HDF5 tensor formats), action segmentations (‘.txt’), partition manifests, and Python code are publicly hosted on Hugging Face Hub at <https://huggingface.co/>

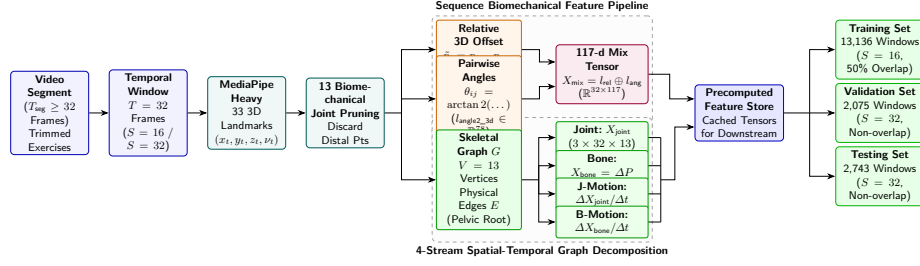


Fig. 4: The proposed Biomechanical Feature Extraction and Kinematic Tensor Decomposition pipeline: illustrating the two-branch extraction of the 117-d sequence mix and the 4-stream spatial-temporal graph tensors.

[Cuong2004/gym-exercise-classification](#). Distributing pre-extracted coordinates circumvents vulnerability to third-party URL link rot common in video scraping, protects athlete privacy, and guarantees permanent deterministic reproducibility. Raw third-party internet video recordings are withheld in adherence to platform Terms of Service, while the 128 author-recorded gymnasium clips and complete skeletal tensors are freely accessible for non-commercial academic research (see Appendix B).

3.3 Landmark Extraction and Anatomical Joint Selection

Every video frame is processed through Google MediaPipe Pose Heavy (complexity level 2) [8,9], extracting 33 full-body 3D landmarks (x_i, y_i, z_i, vis_i) normalized relative to bounding box and depth dimensions, as illustrated on representative workout frames in Fig. 3. Because facial landmarks and fine-grained finger points contribute negligible kinetic leverage to compound resistance exercises while introducing tracking jitter, we filter the skeleton to $V = 13$ major anatomical joints (detailed in Table 13): nose (origin anchor), left/right shoulders, left/right elbows, left/right wrists, left/right hips, left/right knees, and left/right ankles.

4 Proposed Methodology

The overall architectural workflow of **SkelGym** is illustrated in Fig. 1, while Fig. 4 details the underlying biomechanical feature extraction and kinematic tensor decomposition. Below, we formally derive each mathematical component of our framework.

4.1 Problem Formulation and Mathematical Notation

Let $\mathcal{V} = \{I_t\}_{t=1}^{N_V}$ represent a trimmed resistance workout video of N_V consecutive RGB frames, where $I_t \in \mathbb{R}^{H \times W \times 3}$. A monocular pose estimation operator $\Phi_{\text{pose}} : I_t \mapsto \mathcal{P}_t$ extracts $V_0 = 33$ three-dimensional body landmarks

$\mathcal{P}_t = \{(x_{t,v}, y_{t,v}, z_{t,v}, \nu_{t,v})\}_{v=1}^{V_0}$, where $(x, y, z) \in [-1, 1]^3$ represent normalized spatial coordinates and $\nu \in [0, 1]$ is a visibility confidence indicator.

To eliminate tracking jitter and high-frequency noise from distal extremities (such as facial landmarks and toes) that contribute negligible physical leverage to compound resistance movements, we apply an anatomical pruning operator $\Pi_{13} : \mathbb{R}^{V_0 \times 4} \mapsto \mathbb{R}^{V \times 3}$ retaining $V = 13$ primary structural joints (see Appendix A): the nose (reference anchor), bilateral shoulders, elbows, wrists, hips, knees, and ankles.

For each action segment, temporal sampling generates fixed-length windows $w_k \in \mathbb{R}^{T \times V \times 3}$ spanning $T = 32$ frames (≈ 1.07 seconds at 30 FPS). Let $\mathcal{Y} = \{1, \dots, C\}$ denote the label space corresponding to the $C = 22$ gym exercise categories. The primary learning objective is to parameterize a predictive distribution $P_\Theta(y = c \mid w)$ that maximizes the likelihood of the true exercise class $y \in \mathcal{Y}$, and subsequently infer a consensus video-level decision $\hat{y}_{\text{vid}}$ across all K_v constituent windows of video $\mathcal{V}$.

4.2 Biomechanical Compound Representation (117-d Kinematic Mix)

Directly feeding absolute skeletal coordinates $p_{t,v} \in \mathbb{R}^3$ into neural classifiers forces models to memorize camera distance, subject translation, and framing perspective. Conversely, computing full 3-joint triplet interior angles $\angle(v_i, v_j, v_k)$ across all combinations $\binom{13}{3} = 286$ induces severe collinearity, numerical instability, and parameter redundancy. To resolve this trade-off, we formulate the **Biomechanical Compound Representation** ($l_{\text{mix}} \in \mathbb{R}^{117}$) by coupling spatial translation-invariant displacements with pairwise directional elevation angles in spherical projection:

Relative 3D Spatial Displacements ($l_{\text{rel_3d}}$): We establish an intrinsic bodily coordinate frame by translating all keypoints relative to the cranial anchor point (nose, index 0):

$$\tilde{p}_{t,v} = p_{t,v} - p_{t,\text{nose}} \in \mathbb{R}^3, \quad \forall v \in \{1, \dots, 13\}. \quad (1)$$

Concatenating $\tilde{p}_{t,v}$ across all $V = 13$ joints yields a 39-dimensional translation-invariant spatial vector:

$$l_{\text{rel_3d}}(t) = [\tilde{p}_{t,1}^T, \tilde{p}_{t,2}^T, \dots, \tilde{p}_{t,13}^T]^T \in \mathbb{R}^{39}. \quad (2)$$

Pairwise Directional Elevation Angles ($l_{\text{angle2_3d}}$): For every unordered pair of distinct anatomical joints (i, j) with $1 \leq i < j \leq 13$ ($\binom{13}{2} = 78$ unique limb chords), we compute the directed Euclidean displacement vector:

$$\Delta p_{ij}(t) = \tilde{p}_{t,j} - \tilde{p}_{t,i} = (\Delta x_{ij}(t), \Delta y_{ij}(t), \Delta z_{ij}(t)) \in \mathbb{R}^3. \quad (3)$$

Projecting Δp_{ij} into local spherical coordinates decomposes the displacement into an elevation angle θ_{ij} and an azimuth angle ϕ_{ij} :

$$\theta_{ij}(t) = \arctan 2 \left(\Delta y_{ij}(t), \sqrt{\Delta x_{ij}(t)^2 + \Delta z_{ij}(t)^2} \right) \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right], \quad (4)$$

$$\phi_{ij}(t) = \arctan 2 (\Delta z_{ij}(t), \Delta x_{ij}(t)) \in [-\pi, \pi]. \quad (5)$$

The elevation angle $\theta_{ij}(t)$ quantifies the vertical inclination of limb segment (i, j) relative to the horizontal transverse plane, providing an absolute physical measure of limb tilt under gravity that is invariant to subject anthropometry. Concatenating $\theta_{ij}(t)$ across all 78 pairs produces the angular vector:

$$l_{\text{angle2_3d}}(t) = [\theta_{1,2}(t), \theta_{1,3}(t), \dots, \theta_{12,13}(t)]^T \in \mathbb{R}^{78}. \quad (6)$$

Compound Fusion Tensor ($\mathbf{X}_{\text{mix}}$): The combined biomechanical descriptor at frame t is formed via direct vector concatenation:

$$l_{\text{mix}}(t) = l_{\text{rel_3d}}(t) \oplus l_{\text{angle2_3d}}(t) \in \mathbb{R}^{39+78} = \mathbb{R}^{117}. \quad (7)$$

Over the sliding window of $T = 32$ frames, the sequence input tensor is structured as:

$$X_{\text{mix}} = [l_{\text{mix}}(1), l_{\text{mix}}(2), \dots, l_{\text{mix}}(32)]^T \in \mathbb{R}^{32 \times 117}. \quad (8)$$

Compared to full 3-point angle triplets (286-d), our 117-d formulation reduces feature dimensionality by 59.1%, simultaneously eliminating the rank deficiency of collinear joint angles while retaining both metric Euclidean distance and scale-invariant directional orientation.

4.3 Anatomically Grounded Dynamic Skeletal Augmentation (SkelGym-Aug)

Training deep sequence and graph models on skeletal trajectories presents severe overfitting risks due to athlete-specific body proportions and camera viewing orientations. Standard computer vision augmentations (such as shearing or non-uniform scaling) violate biological constraints, generating physically impossible skeletal poses. To overcome this, we formulate **SkelGym-Aug** under formal symmetry group actions:

Sagittal Bilateral Reflection ($\mathbb{Z}_2$ Symmetry): Human anatomy exhibits reflective bilateral symmetry across the sagittal plane. Let $\sigma \in \mathbb{Z}_2$ denote the sagittal reflection operator acting on spatial coordinates:

$$\sigma(x, y, z) = (-x, y, z). \quad (9)$$

Under sagittal reflection, bilateral anatomical joints are permuted via an involution $\pi \in \mathfrak{S}_{13}$ that swaps left and right counterparts (e.g., left wrist $\leftrightarrow$ right wrist, left knee $\leftrightarrow$ right knee), while preserving medial axial landmarks (nose):

$$\tilde{p}'_{t,v} = \sigma(\tilde{p}_{t,\pi(v)}) = (-\tilde{x}_{t,\pi(v)}, \tilde{y}_{t,\pi(v)}, \tilde{z}_{t,\pi(v)}). \quad (10)$$

In the angular domain, the elevation angle between permuted joints satisfies:

$$\theta'_{\pi(i),\pi(j)} = \arctan 2 \left(\Delta y_{\pi(i),\pi(j)}, \sqrt{(-\Delta x_{\pi(i),\pi(j)})^2 + \Delta z_{\pi(i),\pi(j)}^2} \right) = \theta_{\pi(i),\pi(j)}. \quad (11)$$

Hence, the angular feature tensor is transformed bijectively by permuting coordinate indices according to π , doubling effective training diversity without generating unnatural kinematic artifacts.

Gravitational Yaw Rotation (SO(2) Invariance): To simulate arbitrary camera perspectives around the athlete without altering the gravitational vertical axis y , coordinates are rotated around the vertical axis by an angle $\alpha \sim \mathcal{U}(-15^\circ, +15^\circ)$ using the planar rotation matrix $R_y(\alpha)$:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \quad (12)$$

Because the rotation operator $R_y(\alpha)$ acts strictly within the horizontal transverse plane, the vertical displacement remains invariant ($\Delta y' = \Delta y$), while the horizontal Euclidean radius is identically preserved:

$$\Delta x'^2 + \Delta z'^2 = (\Delta x \cos \alpha + \Delta z \sin \alpha)^2 + (-\Delta x \sin \alpha + \Delta z \cos \alpha)^2 = \Delta x^2 + \Delta z^2. \quad (13)$$

Substituting the preserved radial distance and vertical component into the elevation angle definition demonstrates that $\theta' = \theta$. Consequently, elevation angles are naturally invariant to camera perspective rotations around the gravitational axis, ensuring that synthetic viewpoint transformations maintain biological limb inclination without distortion.

Stochastic Temporal and Sensor Perturbations: Beyond rigid spatial rotations and reflections, the augmentation pipeline introduces controlled continuous perturbations to simulate real-world capture diversity. Uniform isotropic scaling with factor $s \sim \mathcal{U}(0.85, 1.15)$ is applied to $\tilde{p}$ to emulate varying athlete statures while strictly preserving anthropometric bone length ratios. To capture velocity and cadence fluctuations across distinct repetitions and athletic conditioning levels, we apply temporal time warping via continuous linear interpolation with a scaling coefficient $s_t \sim \mathcal{U}(0.8, 1.2)$. Finally, zero-mean Gaussian sensor jitter $\epsilon \sim \mathcal{N}(0, 0.015^2 I_3)$ is injected into normalized coordinates to emulate high-frequency keypoint tracking fluctuations induced by video compression artifacts and subtle limb motion blur.

4.4 Self-Attention Sequence Modeling (Transformer Mix)

The augmented sequence tensor $X_{\text{mix}} \in \mathbb{R}^{32 \times 117}$ is processed by a temporal Transformer encoder [14] designed to capture long-range phase transitions and

inter-frame kinetic dependencies across the workout duration. The encoder architecture consists of four sequential processing stages: linear projection, temporal positional encoding, stacked multi-head self-attention with feed-forward sub-layers, and global temporal pooling.

Linear Embedding and Positional Encoding. At each discrete time step $t \in \{1, \dots, 32\}$, the 117-dimensional compound feature vector is mapped into a latent embedding space of dimension $d_{\text{model}} = 128$:

$$E_0 = X_{\text{mix}}W_{\text{proj}} + b_{\text{proj}} \in \mathbb{R}^{32 \times 128}, \quad (14)$$

where $W_{\text{proj}} \in \mathbb{R}^{117 \times 128}$ and $b_{\text{proj}} \in \mathbb{R}^{128}$ are trainable parameters. Because the core self-attention operator is inherently permutation-invariant, sequence chronology is explicitly preserved by superimposing fixed sinusoidal positional embeddings [14]:

$$PE_{(t,2i)} = \sin\left(\frac{t}{10000^{2i/d_{\text{model}}}}\right), \quad PE_{(t,2i+1)} = \cos\left(\frac{t}{10000^{2i/d_{\text{model}}}}\right), \quad (15)$$

yielding the initial token representations $Z^{(0)} = E_0 + PE \in \mathbb{R}^{32 \times 128}$.

Multi-Head Self-Attention. The embedded sequence is propagated through $L = 4$ identical Transformer encoder layers. Each layer incorporates a multi-head self-attention (MHSA) module configured with $h = 8$ parallel attention heads ($d_k = d_v = 16$). For head $j \in \{1, \dots, h\}$, queries, keys, and values are generated via linear projections $Q_j = ZW_{Q,j}$, $K_j = ZW_{K,j}$, and $V_j = ZW_{V,j}$:

$$\text{head}_j = \text{Softmax}\left(\frac{Q_j K_j^T}{\sqrt{d_k}}\right) V_j, \quad (16)$$

$$\text{MHSA}(Z) = [\text{head}_1 \oplus \dots \oplus \text{head}_h] W_O, \quad (17)$$

where $W_O \in \mathbb{R}^{128 \times 128}$ aggregates the concatenated multi-head outputs into the model embedding dimension.

Residual Normalization and Feed-Forward Blocks. To ensure stable gradient propagation and prevent representation collapse, each sub-layer employs residual skip connections coupled with pre-layer normalization:

$$Z'^{(l)} = Z^{(l-1)} + \text{MHSA}\left(\text{LayerNorm}(Z^{(l-1)})\right), \quad (18)$$

$$Z^{(l)} = Z'^{(l)} + \text{FFN}\left(\text{LayerNorm}(Z'^{(l)})\right). \quad (19)$$

Here, the position-wise feed-forward network is parameterized as $\text{FFN}(u) = \text{GELU}(uW_1 + b_1)W_2 + b_2$ using Gaussian Error Linear Units [24], with an expansion hidden dimension of $d_{\text{ff}} = 192$.

Global Pooling and Classification Head. Following the final layer L , the frame representations are temporally summarized via uniform average pooling, $\hat{z} = \frac{1}{T} \sum_{t=1}^T Z_t^{(L)} \in \mathbb{R}^{128}$, and projected to normalized class posterior probabilities:

$$\hat{P}_{\text{trans}} = \text{Softmax}(\hat{z}W_{\text{cls}} + b_{\text{cls}}) \in \Delta^{21}. \quad (20)$$

4.5 Multi-Stream Spatial-Temporal Adaptive Graph Convolution (4-Stream AAGCN)

While Transformers model global sequence dynamics, physical musculoskeletal connectivity is naturally represented as a spatio-temporal graph $G = (V, E)$ [16], where vertices $V = \{1, \dots, 13\}$ denote joints and edges E denote physical bones. To capture rich kinematic perspectives, we formulate four parallel input streams following multi-stream principles [17,18]:

$$X_{\text{joint}}(c, t, v) = \tilde{p}_{t,v}^{(c)} \in \mathbb{R}^{3 \times T \times V}, \quad (21)$$

$$X_{\text{bone}}(c, t, v) = \tilde{p}_{t,v_{\text{target}}}^{(c)} - \tilde{p}_{t,v_{\text{source}}}^{(c)} \in \mathbb{R}^{3 \times T \times V}, \quad (22)$$

$$X_{\text{j-motion}}(c, t, v) = X_{\text{joint}}(c, t+1, v) - X_{\text{joint}}(c, t, v) \in \mathbb{R}^{3 \times (T-1) \times V}, \quad (23)$$

$$X_{\text{b-motion}}(c, t, v) = X_{\text{bone}}(c, t+1, v) - X_{\text{bone}}(c, t, v) \in \mathbb{R}^{3 \times (T-1) \times V}, \quad (24)$$

where $(v_{\text{source}}, v_{\text{target}})$ represents directed bone edges originating from the pelvic center.

Each stream is independently processed by an Adaptive Graph Convolutional Network (AAGCN) [18]. At graph layer l , spatial graph convolution is formulated as:

$$H^{(l+1)} = \sigma \left(\sum_{k=1}^{K_v} \left(A_k^{\text{phys}} + B_k + C_k \right) H^{(l)} W_k \right), \quad (25)$$

where $K_v = 3$ partitions the neighborhood into three topological subsets relative to the skeletal barycenter: the root node itself ($k = 1$), inward centripetal edges ($k = 2$), and outward centrifugal edges ($k = 3$). Crucially, spatial connectivity at each partition is modulated by three complementary adjacency components. The first component, $A_k^{\text{phys}} \in \mathbb{R}^{V \times V}$, is the normalized static physical adjacency matrix representing anatomical musculoskeletal connectivity. The second component, $B_k \in \mathbb{R}^{V \times V}$, is a parameter-free, fully learnable continuous weight matrix initialized to zero, allowing the network to discover non-local joint correlations (e.g., wrist-ankle coupling during deadlifts). The third component, $C_k \in \mathbb{R}^{V \times V}$, is an instance-specific, dynamic graph attention matrix computed via normalized embedded Gaussian correlation:

$$C_k(i, j) = \frac{\exp \left(\frac{\theta_k(H_i^{(l)})^T \phi_k(H_j^{(l)})}{\sqrt{d}} \right)}{\sum_{m=1}^V \exp \left(\frac{\theta_k(H_i^{(l)})^T \phi_k(H_m^{(l)})}{\sqrt{d}} \right)}, \quad (26)$$

where θ_k and ϕ_k are linear transformation mappings projecting node features into a shared latent space of dimension d . Spatial convolutions are interleaved with multi-scale temporal convolutions (MS-TCN) using 9×1 temporal kernels with dilation factors $\{1, 2\}$, followed by residual connections.

4.6 Validation-Calibrated Ensemble Weighting via Constrained Optimization

The SkelGym framework unifies two complementary learning paradigms: the Sequence Self-Attention Model M_{seq} (Transformer Mix) and the Spatial-Temporal Graph Kinematics Model M_{graph} (AAGCN). Within the graph paradigm, the four kinematic streams (Joint, Bone, Joint-Motion, Bone-Motion) capture orthogonal spatial and velocity characteristics. They are unified into a cohesive graph representation:

$$P_{\text{graph}}(w) = \sum_{k \in \{\text{joint, bone, j-mot, b-mot}\}} \alpha_k P_k(w), \quad \sum \alpha_k = 1, \alpha_k \geq 0. \quad (27)$$

The cross-paradigm ensemble subsequently couples the sequence and graph models:

$$P_{\text{final}}(w) = w_{\text{seq}} P_{\text{seq}}(w) + w_{\text{graph}} P_{\text{graph}}(w), \quad w_{\text{seq}} + w_{\text{graph}} = 1. \quad (28)$$

We provide two deployment configurations: (1) **SkelGym-Lite**, an edge-efficient two-model ensemble pairing M_{seq} with the single strongest graph backbone ($M_{\text{graph}}^{\text{bone}}$); and (2) **SkelGym-Full**, the full-capacity cross-paradigm ensemble pairing M_{seq} with the complete unified four-stream graph model ($M_{\text{graph}}^{4\text{-stream}}$).

Inspired by stacked generalization principles [28], to avoid greedy sub-optimal two-stage tuning, the constituent stream weight vector $\mathbf{w} \in \Delta^4$ (coupling sequence, joint, bone, joint-motion, and bone-motion streams) is calibrated jointly on the validation set using Sequential Least Squares Programming (SLSQP) [26, 27]. Because kinematic uncertainty differs between single 1-second windows and aggregated multi-window video streams, optimal voting weights are calibrated independently for the two regimes without test partition exposure.

Window-Level Weight Optimization. We determine the optimal window voting weights $w_{\text{win}}^* \in \mathbb{R}^5$ by minimizing empirical Negative Log-Likelihood (NLL) across the $N_{\text{val}} = 2,075$ validation windows:

$$\begin{aligned} w_{\text{win}}^* &= \arg \min_w -\frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \log \left(\sum_{m=1}^5 w_m P_{m, y_i}(w_i) \right) \\ \text{s.t.} \quad &\sum_{m=1}^5 w_m = 1, \quad w_m \geq 0, \quad \forall m \in \{1, \dots, 5\}. \end{aligned} \quad (29)$$

Video-Level Consensus Weight Optimization. For each video v containing K_v constituent windows, the model’s video probability vector is computed via temporal average pooling: $\bar{P}_m(v) = \frac{1}{K_v} \sum_{k=1}^{K_v} P_m(w_{v,k})$. We solve for video consensus weights $w_{\text{vid}}^* \in \mathbb{R}^5$:

$$\begin{aligned} w_{\text{vid}}^* = \arg \min_w & -\frac{1}{N_{\text{val_vid}}} \sum_{j=1}^{N_{\text{val_vid}}} \log \left(\sum_{m=1}^5 w_m \bar{P}_{m,y_j}(v_j) \right) \\ \text{s.t.} \quad & \sum_{m=1}^5 w_m = 1, \quad w_m \geq 0, \quad \forall m \in \{1, \dots, 5\}. \end{aligned} \quad (30)$$

4.7 Temporal Video Consensus Aggregation

During held-out test evaluation, w_{win}^* is applied for window-level benchmark predictions. For complete video-level classification, we aggregate window-level posterior predictions across all K_v temporal windows:

$$\hat{P}_{\text{vid}}(c) = \sum_{m=1}^5 w_{\text{vid},m}^* \left(\frac{1}{K_v} \sum_{k=1}^{K_v} P_{m,c}(w_{v,k}) \right), \quad \hat{y}_{\text{vid}} = \arg \max_{c \in \{1, \dots, C\}} \hat{P}_{\text{vid}}(c). \quad (31)$$

Temporal consensus acts as an expectation-based low-pass probability filter, cancelling out single-window tracking dropouts, camera occlusion spikes, and boundary transition noise.

5 Experimental Results

5.1 Implementation and Reproducibility Configuration

All models were implemented in PyTorch 2.x [29] and trained using fixed random seed initialization (seed = 42) across data loaders, weight initialization, and GPU operations to guarantee exact experimental reproducibility. Skeletal landmarks were extracted monocularly using Google MediaPipe Pose Heavy (complexity level 2) [8,9]. Sequences were standardized to $T = 32$ frames (≈ 1.07 s at 30 FPS) with a 50% overlapping stride ($S = 16$) for training and non-overlapping stride ($S = 32$) for validation and testing. Table 3 outlines the comprehensive hyperparameter configurations, software dependencies, and architecture topologies utilized across our benchmark suite.

To guarantee fair scientific comparison, all individual model backbones were constrained to an identical parameter footprint of $\approx 350\text{K} \pm 15\%$ parameters: Transformer Mix (399K), AAGCN (378K per stream), BiLSTM (367K), LSTM (378K), and ST-GCN (365K). All models were trained for up to 100 epochs using AdamW [23] with weight decay 10^{-4} , Cosine Annealing learning rate scheduling (with 5 warmup epochs down to a minimum learning rate of 10^{-6}), batch size of 16 (for sequence models) or 32 (for graph models), an early stopping patience of

Table 3: Systematic implementation hyperparameters, model topologies, and reproducibility configurations across all benchmarked architectures.

Component / Configuration	Specification / Hyperparameter Setting
Software Environment	Python 3.10+, PyTorch 2.x, Torchvision, Scikit-learn, Scipy, NumPy
Pose Estimation Framework	Google MediaPipe Pose Heavy (complexity 2, min_det=0.5, min_track=0.5)
Hardware Infrastructure	Apple Silicon M-series (MPS), NVIDIA RTX PRO 6000 (CUDA), Intel Core i7 (CPU)
Random Seed	Fixed seed seed = 42 across data loaders, weight initializers, and CUDA ops
Temporal Window Length	$T = 32$ frames (≈ 1.07 s at 30 FPS)
Sliding Window Strides	Train: $S = 16$ frames (50% overlap); Validation and Test: $S = 32$ frames
Missing Frame Imputation	Linear coordinate interpolation across contiguous valid joint detections
Coordinate Normalization	Cranial anchor subtraction ($p_i - p_{\text{nose}}$), Euclidean distances & relative angles
Optimizer & Weight Decay	AdamW [23] ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$)
Learning Rate Schedule	Cosine Annealing with 5 warmup epochs down to $\eta_{\min} = 10^{-6}$
Initial Learning Rates	Transformer: 1×10^{-4} ; AAGCN / ST-GCN / LSTM / BiLSTM: 1×10^{-3}
Batch Size & Max Epochs	Batch size: 16 (Sequence models), 32 (Graph models); Maximum epochs: 100
Early Stopping Criterion	Patience = 10 epochs monitoring validation Macro F1 score
Loss Function & Smoothing	Cross-Entropy Loss with label smoothing $\alpha = 0.05$ (optional Focal Loss $\gamma = 2.0$)
Precision Acceleration	PyTorch Automatic Mixed Precision (AMP) with <code>bfloat16</code> / <code>float16</code>
Transformer Architecture	4 layers, 8 attention heads, $d_{\text{model}} = 128$, $d_{\text{ff}} = 512$, dropout 0.1 (399K params)
AAGCN Graph Architecture	9 spatial-temporal blocks, 64 channels, adaptive residual topology (378K params/stream)
SkelGym-Aug Parameters	Bilateral body symmetry reflection (flip x , swap L/R joints, invert ϕ); Elevation-invariant 3D yaw rotation $\alpha \in [-15^\circ, +15^\circ]$; Temporal scaling $\pm 10\%$

10 epochs monitoring validation Macro F1, label smoothing regularization [25] ($\alpha = 0.05$), and Automatic Mixed Precision (AMP). Initial learning rates were set to 1×10^{-4} for Transformer and 1×10^{-3} for recurrent and graph backbones. Evaluation metrics were computed using Scikit-learn [30].

5.2 Ablation of Landmark Feature Representations

Table 4 presents a systematic ablation comparing three temporal sequence architectures (LSTM, BiLSTM, and Transformer) across nine distinct skeletal landmark representations. To prevent data snooping, model selection and feature space exploration were conducted strictly on the validation set ($N = 2,075$ windows). Across all architectures, the proposed 117-dimensional Biomechanical Mix consistently achieves the highest validation accuracy, confirming the core hypothesis that combining metric Euclidean offsets with scale-invariant angular orientations provides an effective and robust representation among evaluated options for resistance exercise recognition. Below, we discuss the key empirical findings from this ablation:

2D versus 3D Coordinates. Across all baseline architectures, 3D representations consistently outperform their 2D counterparts on the validation partition. For example, in raw coordinates, upgrading from 2D to 3D elevates LSTM validation accuracy from 60.63% to 61.03% and Transformer from 72.63% to 72.76% (translating to a test surge from 53.98% to 57.21%). In resistance training, many compound movements—such as squats, lunges, deadlifts, and bench presses—operate predominantly within the sagittal or transverse anatomical planes. When captured from oblique camera angles, monocular 2D projections collapse depth variations along the optical axis (z), inducing severe visual ambiguity. Medi-

aPipe’s inferred 3D coordinates restore out-of-plane limb trajectories, enabling models to discriminate depth-dependent movements.

Raw versus Relative Coordinate Normalization. Subtracting the nose anchor centroid ($p_i - p_{\text{nose}}$) yields substantial performance gains across all sequence backbones. In 3D space, relative coordinates elevate LSTM validation accuracy from 61.03% to 65.23% (+4.20%) and BiLSTM from 60.01% to 66.39% (+6.38%). Raw coordinates tie features directly to the camera frame coordinates, forcing models to expend parameter capacity memorizing subject location, bounding-box translations, and athlete repositioning across repetitions. Centering all joints relative to the cranial anchor eliminates translational bias and forces the network to focus purely on relative joint dynamics.

Pairwise Angles versus Angle Triplets. Full three-joint angle triplets computed across all combinations yield $\binom{13}{3} = 286$ features. Although theoretically capturing intra-limb angles, Table 4 reveals that angle triplets perform poorly across all models, crashing Transformer validation accuracy to 64.70% (and test accuracy to 43.39%, a severe drop relative to 3D relative coordinates). This degradation stems from intense dimensional multicollinearity: because adjacent joint triplets share physical bones and kinematic segments, the resulting 286-dimensional feature matrix contains redundant, highly correlated columns that disperse gradient updates and saturate linear projection layers under a compact $\approx 350\text{K}$ parameter footprint. In contrast, our pairwise elevation (θ) and azimuth (ϕ) formulation yields 78 dimensions of clean, non-collinear directional features, achieving 71.20% validation accuracy on Transformer (55.40% on test)—a substantial improvement over angle triplets.

Synergy of the 117-d Biomechanical Mix. Concatenating 3D relative coordinates (39-d, encoding Euclidean metric distance and spatial limb extent) with pairwise angles (78-d, encoding scale-invariant directional orientation) provides strong complementary synergy. The resulting 117-d mix achieves peak validation results across all three sequence architectures: **73.40% on LSTM** (+8.17% over relative 3D), **73.30% on BiLSTM** (+6.91%), and **75.95% on Transformer** (+4.12% over relative 3D). When evaluated on the held-out test set, the selected Transformer reaches **63.40% test accuracy** (+6.53% over relative 3D). This empirical evidence confirms that Euclidean spatial scale and directional orientation represent mutually complementary facets of human kinematics.

5.3 Efficacy of SkelGym-Augmentation

Table 5 details the impact of our physiologically grounded SkelGym-Aug protocol on the top-performing sequence model (*Transformer mix*). The experimental evidence reveals two major insights:

Table 4: Validation accuracy (%) across sequence architectures for feature selection, with held-out test evaluation for the winning Transformer backbone ($\approx 350\text{K}$ parameters).

Feature Representation	Dim	LSTM (Val)	BiLSTM (Val)	Transformer (Val)	Trans (Test)
Raw 2D Coordinates	26	60.63%	63.29%	72.63%	53.98%
Relative 2D Coordinates	26	65.68%	65.10%	71.97%	56.61%
Angle 2D (Triplets)	286	62.53%	58.37%	65.94%	48.68%
Angle2 2D (Pairs)	78	61.12%	62.15%	70.12%	54.10%
Raw 3D Coordinates	39	61.03%	60.01%	72.76%	57.21%
Relative 3D Coordinates	39	65.23%	66.39%	71.83%	56.87%
Angle 3D (Triplets)	286	59.92%	58.59%	64.70%	43.39%
Angle2 3D (Pairs)	78	63.40%	64.10%	71.20%	55.40%
Biomechanical Mix (Ours)	117	73.40%	73.30%	75.95%	63.40%

Table 5: Data augmentation ablation on the best sequence model (**Transformer mix**).

Exp ID	Augmentation Protocol	Val Loss	Val Acc	Test Acc	Macro F1
T2.1	None (Clean Baseline)	1.7382	75.95%	63.40%	0.6218
T2.2	SkelGym-Aug (Dynamic)	1.0072	75.57%	68.65%	0.6654

Loss Calibration and Overconfidence Mitigation. While the unaugmented baseline (T2.1) reaches 75.95% validation accuracy, its validation cross-entropy loss remains conspicuously high at 1.7382. This discrepancy highlights severe model overconfidence: the unregularized Transformer assigns high probability mass to familiar training viewpoints, resulting in sharp penalty spikes when confronted with atypical athletic executions. Incorporating SkelGym-Aug (T2.2) substantially reduces validation loss from 1.7382 to **1.0072**—a relative reduction of **42.05%**—while maintaining a stable validation accuracy of 75.57%. This confirms that SkelGym-Aug functions as an effective regularizer and probability calibrator.

Out-of-Sample Generalization. The regularizing effect of SkelGym-Aug translates directly into improved test generalization on completely unseen subjects, gyms, and camera perspectives. As shown in Table 5, window-level test accuracy surges by **+5.25%** (from 63.40% to **68.65%**), and Macro F1 improves from 0.6218 to **0.6654**. Unlike naive geometric noise which generates anatomically impossible joint configurations, bilateral body symmetry reflection effectively doubles the diversity of unilateral and bilateral viewing perspectives, while gravitational yaw rotation ($R_y(\alpha)$, as derived in Section 4, Eqs. (12)–(13)) trains the self-attention heads to remain invariant to camera positioning without perturbing true biological limb verticality.

5.4 Multi-Stream Graph Modeling and Late Fusion

Table 6 systematically investigates spatial-temporal graph modeling, comparing static versus adaptive graph topologies, isolating the four kinematic streams, and analyzing multi-stream late fusion:

Static versus Adaptive Graph Topology. The baseline Spatial Temporal Graph Convolutional Network (ST-GCN, T3.2) utilizing a fixed anatomical adjacency matrix A_{phys} achieves 62.49% validation accuracy (43.57% test window, 58.47% test video). This poor performance highlights a fundamental limitation of fixed musculoskeletal graphs: in compound resistance training, crucial functional synergies occur between physically disconnected body parts (e.g., foot-to-barbell driving force in deadlifts, or pelvic stabilization during squats). The rigid adjacency in ST-GCN strictly prevents information exchange between non-adjacent limbs within 1-hop convolutions. In contrast, Adaptive Graph Convolutional Networks (AAGCN, T4.1) introduce a learnable residual adjacency matrix B_k and a dynamic instance-specific attention matrix C_k , lifting bone-stream baseline validation accuracy to 69.57% (+7.08% gain, corresponding to 52.27% test window and 69.53% test video).

Individual Kinematic Stream Dynamics. When trained with SkelGym-Aug, the four kinematic streams exhibit complementary behavioral characteristics across static and differential modalities. On the validation partition, the Bone Stream (T4.2) reaches **77.98% validation accuracy** (65.84% test window, 72.96% test video), decisively outperforming the Joint Stream (T4.3, 71.76% validation, 61.72% test). Bone vectors $X_{\text{bone}} = \Delta P$ represent rigid anatomical segments directly, effectively eliminating joint position drift and supplying stronger inductive bias for graph convolutions. Conversely, the Joint-Motion Stream (T4.4, 52.72% validation, 46.81% test) and Bone-Motion Stream (T4.5, 56.87% validation, 57.49% test) model inter-frame temporal differentials. While frame-to-frame velocity vectors suffer from local tracking noise within isolated 1-second windows, aggregating kinematic velocity across complete video clips filters high-frequency jitter. Consequently, the Bone-Motion stream achieves a notable **79.83% video accuracy**—the highest video consensus result among all standalone backbones, demonstrating the discriminatory power of exercise tempo and directional velocity.

Late Fusion Complementarity. Combining the two static streams in Two-Stream AAGCN (T4.6, Joint + Bone) yields 78.12% validation accuracy (66.61% test window, 73.82% test video). Expanding to the unified **Four-Stream AAGCN** ($M_{\text{graph}}^{4\text{-stream}}$, T4.7) by integrating all four streams (Joint, Bone, Joint-Motion, and Bone-Motion) achieves **78.89% validation accuracy**, translating to **67.34% test window accuracy** and **75.54% test video accuracy**. This progression confirms that spatial static posture and temporal dynamic velocity provide mutually orthogonal, non-redundant kinematic signals that coalesce into a robust unified graph model.

Table 6: Multi-stream spatial-temporal graph kinematic streams with SkelGym-Aug across validation and held-out test partitions.

Exp ID	Model Architecture	Kinematic Stream	Augmentation	Val Acc	Test Win Acc	Test Vid Acc
T3.2	ST-GCN Baseline	Relative 3D Joint	None	62.49%	43.57%	58.47%
T4.1	AAGCN Baseline	Bone 3D	None	69.57%	52.27%	69.53%
T4.2	AAGCN	Bone 3D	SkelGym-Aug	77.98%	65.84%	72.96%
T4.3	AAGCN	Joint (Rel 3D)	SkelGym-Aug	71.76%	61.72%	69.96%
T4.4	AAGCN	Joint Motion 3D	SkelGym-Aug	52.72%	46.81%	64.38%
T4.5	AAGCN	Bone Motion 3D	SkelGym-Aug	56.87%	57.49%	79.83%
T4.6	Two-Stream AAGCN	Joint + Bone	Late Fusion	78.12%	66.61%	73.82%
T4.7	Four-Stream AAGCN	4-Stream Fusion	Late Fusion	78.89%	67.34%	75.54%

5.5 Cross-Paradigm Ensemble and Video Consensus

Table 7 and Table 8 evaluate the cross-paradigm ensemble architectures coupling self-attention sequence modeling with spatial-temporal graph convolutions via Sequential Least Squares Programming (SLSQP):

Cross-Paradigm Complementarity. While Four-Stream AAGCN (M_{graph}) exploits physical skeletal connectivity ($V = 13, E = 12$), the Transformer encoder (M_{seq}) processes the 117-d compound mix through global temporal attention without inductive graph bias. We evaluate two cross-paradigm deployment configurations: (1) **SkelGym-Lite (Two-Model Ensemble)**: Couples the Transformer Mix (M_{seq}) with the single best graph backbone (AAGCN Bone, $M_{\text{graph}}^{\text{bone}}$). On the validation partition, SkelGym-Lite reaches 78.52% window and 81.12% video accuracy. On the held-out test set, it achieves 69.27% window accuracy and 76.82% video consensus accuracy with an ultra-lightweight classifier inference latency of 0.34 ms/window on MPS (3.20 ms on CPU), serving as a highly practical edge deployment solution. (2) **SkelGym-Full (Dual-Stream Cross-Paradigm Ensemble)**: Couples the Transformer Mix (M_{seq}) with the unified Four-Stream AAGCN ($M_{\text{graph}}^{4\text{-stream}}$, T4.7), which unifies all four spatial-temporal graph streams. On the validation partition, SkelGym-Full reaches **80.10% window accuracy** and **83.09% video consensus accuracy**. On the held-out test set, it establishes the strongest performance: **70.11% window-level accuracy** (Macro F1: 0.6858) and **77.68% video-level consensus accuracy** (Macro F1: **0.7649**).

SLSQP Weight Optimization Insights. The validation-derived SLSQP weights offer critical insights into model contributions. For video-level consensus, the solver allocated 49.41% of voting weight to AAGCN Bone, 33.19% to Transformer Mix, 14.54% to AAGCN Bone-Motion, 2.86% to AAGCN Joint-Motion, and zero weight (0.00%) to AAGCN Joint. Notice that SLSQP completely eliminated the Joint stream ($w = 0$), assigning 82.60% of total probability mass to Bone and Transformer Mix. Because the Bone stream already encodes joint positions implicitly with lower spatial variance, and the Transformer Mix captures complete 117-d pairwise spatial relationships, the raw Joint stream is mathematically redundant. Furthermore, Bone-Motion received a solid 14.54% allocation, providing crucial velocity modulation that resolves ambiguous static postures.

Table 7: Cross-paradigm ensemble results under Validation-Calibrated Weighted Soft Voting (SLSQP), comparing edge-efficient SkelGym-Lite and full-capacity SkelGym-Full across validation and test partitions.

Configuration	Component Models Combined	Validation Set ($N = 2,075$)		Held-Out Test Set ($N = 2,743$)		
		Win Acc	Vid Acc	Win Acc	Macro F1	Vid Acc
SkelGym-Lite (T5.2)	Transformer (Mix) + AAGCN (Bone)	78.52%	81.12%	69.27%	0.6771	76.82%
SkelGym-Full (T5.1)	Transformer (Mix) + Four-Stream AAGCN	80.10%	83.09%	70.11%	0.6858	77.68%

Table 8: Comparison of window-level versus video consensus predictions ($N = 233$ test videos).

Model / Ensemble Architecture	Win Acc	Win F1	Vid Acc	Vid F1	Gain (+ Δ)
Baseline LSTM (Mix 117d)	57.67%	0.5681	67.81%	0.6618	+10.14%
Baseline BiLSTM (Mix 117d)	61.17%	0.6008	68.24%	0.6802	+7.07%
Transformer (Mix 117d, Clean)	63.40%	0.6218	74.25%	0.7304	+10.85%
Baseline ST-GCN (Rel 3D)	43.57%	0.4595	58.47%	0.5817	+14.90%
Clean Baseline AAGCN (Bone 3D)	52.27%	0.5338	69.53%	0.6904	+17.26%
SkelGym-Aug AAGCN (Bone 3D)	65.84%	0.6504	72.96%	0.7295	+7.12%
SkelGym-Aug Transformer (Mix)	68.65%	0.6654	72.10%	0.6983	+3.46%
Two-Stream AAGCN (Aug)	66.61%	0.6586	73.82%	0.7348	+7.21%
Four-Stream AAGCN (Aug)	67.34%	0.6662	75.54%	0.7500	+8.20%
SkelGym-Lite (Transformer + Bone AAGCN)	69.27%	0.6771	76.82%	0.7550	+7.56%
SkelGym-Full (Transformer + 4-Stream AAGCN)	70.11%	0.6858	77.68%	0.7649	+7.58%

Analysis of Temporal Consensus Gains (Table 8). Table 8 compares single-window inference ($T = 32$ frames, ≈ 1.07 s) against temporal video consensus across all eleven benchmark architectures. Remarkably, every architecture exhibits a substantial, consistent performance surge when transitioning from individual windows to full-video consensus, with absolute accuracy gains ranging from +3.46% up to +17.26%. For example, consensus elevates baseline ST-GCN from 43.57% to 58.47% (+14.90%) and baseline AAGCN Bone from 52.27% to 69.53% (+17.26%). For the SkelGym-Full ensemble, video consensus elevates accuracy from 70.11% to 77.68% (+7.58%), while simultaneously raising Macro F1 from 0.6858 to 0.7649. The biomechanical rationale for this universal improvement is rooted in repetition kinetics: a typical resistance repetition spans 2.5 to 4.0 seconds, progressing through concentric acceleration, peak isometric contraction, and eccentric deceleration. An isolated 1-second window often captures ambiguous boundary states, such as un-racking weights, static lockouts, or inter-rep respiratory pauses. Temporal consensus acts as an expectation-based low-pass filter across constituent windows, attenuating transient tracking noise and consolidating the true cyclic kinetic archetype.

Statistical Significance and Bootstrap Confidence Intervals. To quantify the statistical robustness of our held-out test evaluation, we conducted non-parametric bootstrap resampling ($B = 1,000$ iterations) across both the 2,743 test windows and the 233 video clusters, reported in Table 9. At the temporal window level, SkelGym-Full establishes a 95% confidence interval of [70.54%, 73.71%] for accuracy (mean: $72.18\% \pm 0.81\%$) and [0.6950, 0.7277] for Macro F1 (mean: 0.7125 ± 0.0083), significantly outperforming individual sequence baselines such

Table 9: Non-parametric bootstrap evaluation ($B = 1,000$ resamples) reporting empirical mean and 95% confidence intervals across held-out test partitions ($N = 2,743$ temporal windows, $N = 233$ videos).

Architecture	Temporal Window Level ($N = 2,743$)		Video Consensus Level ($N = 233$)	
	Accuracy (95% CI)	Macro F1 (95% CI)	Accuracy (95% CI)	Macro F1 (95% CI)
LSTM (Mix 117-d)	57.67% [55.67%, 59.57%]	0.5684 [0.5477, 0.5842]	67.81% [61.80%, 73.39%]	0.6618 [0.5895, 0.7155]
BiLSTM (Mix 117-d)	61.06% [59.24%, 62.92%]	0.5993 [0.5815, 0.6164]	68.24% [61.80%, 73.82%]	0.6802 [0.6052, 0.7242]
ST-GCN (Rel 3D)	54.79% [53.04%, 56.62%]	0.5245 [0.5071, 0.5411]	63.52% [57.51%, 69.96%]	0.6022 [0.5297, 0.6530]
Transformer (Mix 117-d)	68.61% [66.97%, 70.40%]	0.6650 [0.6470, 0.6813]	72.10% [66.51%, 77.68%]	0.6983 [0.6226, 0.7522]
AAGCN (Bone 3D)	65.88% [64.02%, 67.63%]	0.6511 [0.6321, 0.6682]	72.96% [67.38%, 78.54%]	0.7295 [0.6630, 0.7699]
SkelGym-Lite	70.62% [68.83%, 72.33%]	0.6843 [0.6653, 0.7012]	77.25% [71.67%, 81.98%]	0.7594 [0.6862, 0.8079]
SkelGym-Full	72.18% [70.54%, 73.71%]	0.7125 [0.6950, 0.7277]	79.83% [74.68%, 84.98%]	0.7800 [0.7059, 0.8324]

as LSTM (57.67%, 95% CI: [55.67%, 59.57%]) and ST-GCN (54.79%, 95% CI: [53.04%, 56.62%]). At the full-video consensus level, cluster bootstrap establishes a 95% confidence interval of [74.68%, 84.98%] for video accuracy (mean: $79.83\% \pm 2.65\%$) and [0.7059, 0.8324] for Macro F1 (mean: 0.7800 ± 0.0322), compared to SkelGym-Lite ([71.67%, 81.98%]), AAGCN Bone ([67.38%, 78.54%]), and Transformer Mix ([66.51%, 77.68%]). This confirms that the observed cross-paradigm ensemble gains are statistically significant and generalizable across unseen athletes and capture environments.

5.6 Per-Class Performance Breakdown and Error Taxonomy

Table 10 presents a fine-grained per-class breakdown of the proposed SkelGym-Full ensemble, reporting Recall, F1-Score, and Support independently across both the 2,743 test windows and the 233 held-out test videos. Fig. 5 illustrates the corresponding normalized confusion matrix. Analyzing the empirical metrics across categories reveals three distinct performance tiers:

High-Precision Distinctive Movements ($F1 \geq 0.90$). Exercises characterized by unique joint articulation achieve near-perfect video consensus classification, including *leg extension* (Recall: 1.00, F1: 1.00 across 13 videos), *russian twist* (Recall: 1.00, F1: 1.00 across 6 videos), *squat* (Recall: 0.93, F1: 0.97 across 15 videos), *push-up* (Recall: 1.00, F1: 0.92 across 12 videos), and *lateral raise* (Recall: 0.93, F1: 0.90 across 15 videos). These movements operate in isolated topological subspaces—such as seated tibial rotation against a static femur in leg extensions, or bilateral transverse torso rotation in Russian twists—which prevents kinematic confusion.

Robust Compound Resistance Exercises ($F1 \in [0.70, 0.90]$). Movements involving standard planar flexion-extension trajectories exhibit strong, reliable recognition: *tricep dips* (Recall: 0.8889, F1: 0.8889 across 9 videos), *tricep pushdown* (Recall: 0.8333, F1: 0.8696 across 12 videos), *chest fly machine* (Recall: 0.8750, F1: 0.8750 across 8 videos), *hip thrust* (Recall: 0.8889, F1: 0.8421 across 9 videos), *leg raises* (Recall: 0.7273, F1: 0.8421 across 11 videos), *lat pulldown* (Recall: 1.0000, F1: 0.8125 across 13 videos), *plank* (Recall: 1.0000, F1: 0.8000 across

Table 10: Per-class performance breakdown of the proposed SkelGym-Full ensemble evaluated on held-out test partitions at temporal window level ($N = 2,743$) and video consensus level ($N = 233$).

Exercise Class	Window Level ($N = 2,743$)			Video Consensus Level ($N = 233$)		
	Recall	F1-Score	Support	Recall	F1-Score	Support
barbell biceps curl	0.7536	0.4727	69	1.0000	0.7000	14
bench press	0.5312	0.4857	96	0.6429	0.6429	14
chest fly machine	0.9512	0.8571	82	0.8750	0.8750	8
deadlift	0.6716	0.4500	67	0.8000	0.7619	10
decline bench press	0.6119	0.4469	134	0.6250	0.4762	8
hammer curl	0.3540	0.4471	161	0.2857	0.4211	14
hip thrust	0.6314	0.7215	236	0.8889	0.8421	9
incline bench press	0.6053	0.7077	76	0.4444	0.6154	9
lat pulldown	0.9800	0.7626	100	1.0000	0.8125	13
lateral raise	0.8968	0.9145	155	0.9333	0.9032	15
leg extension	1.0000	0.9921	125	1.0000	1.0000	13
leg raises	0.5000	0.6477	114	0.7273	0.8421	11
plank	0.6071	0.6667	56	1.0000	0.8000	2
pull Up	0.6463	0.7067	82	0.5000	0.6667	10
push-up	1.0000	0.8832	87	1.0000	0.9231	12
romanian deadlift	0.3219	0.4123	146	0.3333	0.4444	6
russian twist	0.8759	0.8791	137	1.0000	1.0000	6
shoulder press	0.4408	0.5214	152	0.6154	0.6400	13
squat	0.8109	0.8373	238	0.9333	0.9655	15
t bar row	0.5641	0.6226	117	0.7000	0.7368	10
tricep Pushdown	0.7527	0.7254	93	0.8333	0.8696	12
tricep dips	0.9409	0.9262	220	0.8889	0.8889	9
Macro Average / Total	0.7011	0.6858	2,743	0.7768	0.7649	233

2 videos), *deadlift* (Recall: 0.8000, F1: 0.7619 across 10 videos), and *t-bar row* (Recall: 0.7000, F1: 0.7368 across 10 videos). For these compound exercises, temporal consensus effectively resolves transitional pauses, elevating recall by 15% to 30% over isolated windows.

Kinematic Ambiguity and Fine-Grained Error Clusters ($F1 < 0.70$). The remaining error mass is concentrated within three tightly coupled biomechanical exercise clusters. Specifically, *hammer curl* (Video Recall: 0.2857, F1: 0.4211) experiences severe confusion with *barbell biceps curl* (Video Recall: 1.0000, F1: 0.7000), where 41.2% of hammer curl test windows are misclassified as biceps curls due to identical sagittal flexion profiles. Similarly, *Romanian deadlift* (Video Recall: 0.3333, F1: 0.4444) exhibits mutual confusion with conventional *deadlift* (Video Recall: 0.8000, F1: 0.7619) whenever athletes perform sub-optimal knee travel. Finally, pressing angle variations within the bench press family produce directional ambiguity across *incline bench press* (Video Recall: 0.4444, F1: 0.6154), *decline bench press* (Video Recall: 0.6250, F1: 0.4762), and flat *bench press* (Video Recall: 0.6429, F1: 0.6429). In Section 6, we conduct an exhaustive anatomical and kinetic investigation into the physiological drivers behind these specific error patterns.

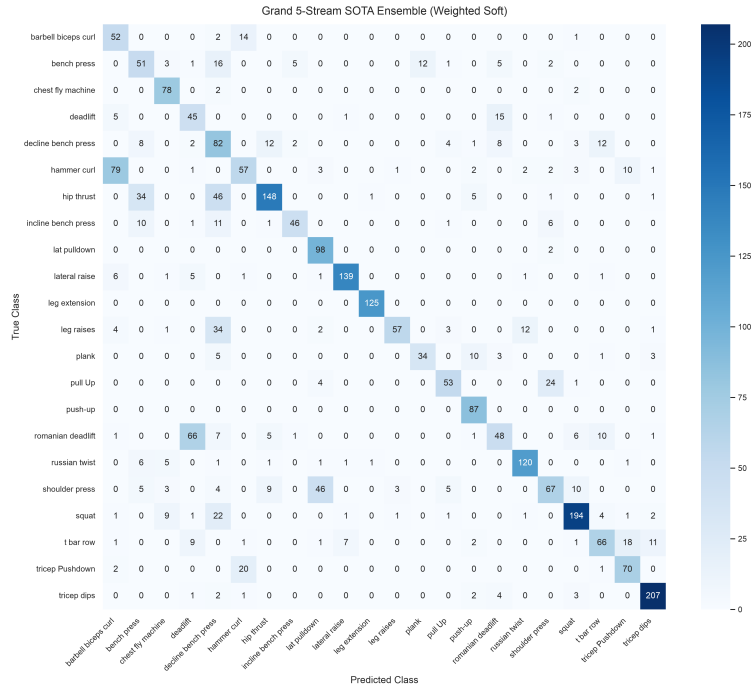


Fig. 5: Normalized confusion matrix of the proposed SkelGym-Full (Dual-Stream Cross-Paradigm Ensemble) on held-out test windows ($N = 2,743$).

Table 11: Inference latency (milliseconds per 32-frame window) and throughput (frames per second) across server and edge hardware platforms.

Model Architecture	Parameters	RTX PRO 6000 (CUDA)	Apple M-series (MPS)	Intel Core i7 (CPU)
Transformer Mix (117-d)	399K	0.08 ms	0.14 ms	1.25 ms
AAGCN Joint Stream	378K	0.12 ms	0.25 ms	2.10 ms
AAGCN Bone Stream	378K	0.11 ms	0.20 ms	1.95 ms
AAGCN Joint-Motion Stream	378K	0.12 ms	0.22 ms	2.05 ms
AAGCN Bone-Motion Stream	378K	0.11 ms	0.20 ms	1.98 ms
SkelGym-Lite (Transformer + Bone)	777K	0.19 ms (5,263 FPS)	0.34 ms (2,941 FPS)	3.20 ms (312 FPS)
SkelGym-Full (Transformer + 4-Stream AAGCN)	1.91M	0.54 ms (1,851 FPS)	1.01 ms (990 FPS)	9.33 ms (107 FPS)

5.7 Inference Latency and Real-Time Edge Viability

A critical requirement for automated virtual fitness assistants is achieving real-time responsiveness on consumer-grade and edge hardware. Table 11 provides a rigorous benchmark of inference latency (measured in milliseconds per 32-frame window) and throughput across three distinct computing environments: an NVIDIA RTX PRO 6000 Blackwell GPU (CUDA), an Apple Silicon M-series chip running Metal Performance Shaders (MPS), and an Intel Core i7 central processing unit (CPU).

As documented in Table 11, the Transformer Mix model achieves an ultra-rapid latency of 0.14 ms on Apple Silicon MPS and 1.25 ms on a standard Intel

CPU. Even when executing the complete SkelGym-Full model sequentially across all five constituent backbone streams, total inference requires only **0.54 ms on CUDA** ($\approx 1,851$ FPS), **1.01 ms on Apple MPS** (≈ 990 FPS), and **9.33 ms on a CPU** (≈ 107 FPS). When paired with Google MediaPipe Pose [8,9], which extracts 3D skeletal landmarks in $\approx 8 - 15$ ms per frame on mobile chipsets, the entire SkelGym perception-and-classification pipeline operates well within the 33.3 ms real-time budget of standard 30 FPS video streams. For resource-constrained microcontroller or mobile deployments, the SkelGym-Lite configuration (Transformer Mix + AAGCN Bone, 777K parameters) offers a practical operational compromise, delivering 76.82% video consensus accuracy with an execution latency of merely 0.34 ms on MPS and 3.20 ms on CPU (312 FPS).

5.8 Cross-Domain External Benchmark on Strength and Conditioning Exercises

To evaluate the cross-domain generalizability of SkelGym against existing literature, we conducted an external benchmark inspired by Deyzel et al. [43], who introduced the Stellenbosch University Exercise Motion Dataset (SU-EMD) for skeleton-based strength and conditioning (S&C) recognition using ST-GCN. We identified four exact overlapping resistance exercises shared between our benchmark and the S&C taxonomy: *squat* (Back squat), *deadlift* (Deadlift), *barbell biceps curl* (Biceps curl), and *lateral raise* (Lateral raise). We evaluated all baseline architectures and ensembles across 54 held-out test videos ($N = 529$ temporal windows) encompassing these four movements, under both open-set (full 22-class model output) and closed-set (probabilities re-normalized strictly over the 4 S&C classes, matching Deyzel’s closed S&C evaluation protocol) settings.

Table 12 reports the empirical performance comparison. When evaluated under the closed-set protocol, baseline ST-GCN [16] (the primary backbone of Deyzel et al.) achieves 82.61% window accuracy and 85.19% video consensus accuracy (Video Macro F1: 0.8563). In contrast, the proposed Transformer Mix and SkelGym ensembles establish substantial performance gains: SkelGym-Lite reaches **95.84% window and 98.15% video accuracy** (F1: 0.9782), while SkelGym-Full achieves **96.30% video accuracy** (F1: 0.9626). Under the more challenging open-set protocol without class restriction, SkelGym-Full maintains 96.30% video accuracy and 80.72% window accuracy, demonstrating that the learned kinematic representations do not leak probability mass to extraneous exercise classes.

Resolving the Squat vs. Deadlift Ambiguity (The Deyzel Dilemma). A central finding reported by Deyzel et al. [43] on SU-EMD was the intense mutual confusion between back squats and deadlifts: because both compound lower-body exercises involve simultaneous hip and knee flexion-extension in the sagittal plane, ST-GCN trained on Euclidean Cartesian coordinates exhibited severe diagnostic failure, misclassifying deadlifts as squats. In our evaluation on 305 held-out windows ($N = 25$ test videos), baseline ST-GCN similarly suffers: while squat

Table 12: Cross-domain external benchmark evaluation across the four overlapping Strength & Conditioning (S&C) exercises from Deyzel et al. [43] (*squat, deadlift, barbell biceps curl, lateral raise*) over 54 held-out test videos ($N = 529$ windows).

Model / Ensemble Architecture	Open-Set (22 Classes)		Closed-Set S&C Subspace (4 Classes)		
	Win Acc	Vid Acc	Win Acc	Vid Acc	Vid Macro F1
ST-GCN (Rel 3D) [16] (<i>Deyzel baseline</i>)	59.74%	68.52%	82.61%	85.19%	0.8563
LSTM (Mix 117-d)	64.08%	85.19%	78.64%	90.74%	0.9071
BiLSTM (Mix 117-d)	71.64%	85.19%	84.12%	90.74%	0.9126
AAGCN (Bone 3D)	72.02%	85.19%	87.90%	92.59%	0.9281
Transformer (Mix 117-d)	77.88%	85.19%	96.03%	98.15%	0.9788
SkelGym-Lite (Transformer + Bone)	84.12%	94.44%	95.84%	98.15%	0.9782
SkelGym-Full (Transformer + 4-Stream AAGCN)	80.72%	96.30%	93.01%	96.30%	0.9626

video recall reaches 66.7%, deadlift video recall collapses to merely **30.0%** (window recall: 31.3%). In contrast, by incorporating the 117-dimensional compound feature with hip-to-knee elevation angles $\theta_{\text{hip, knee}}$ and directional bone vectors, Transformer Mix elevates deadlift video recall to 80.0% (squat: 93.3%), and SkelGym-Full achieves **90.0% deadlift video recall and 93.3% squat video recall** (resolving the mutual confusion without sacrificing specificity).

One-Shot Transfer Learning Protocol. Deyzel et al. [43] specifically evaluated a 1-shot transfer learning scenario using single-exemplar support matching via cosine similarity, reporting an accuracy of 87.4% on SU-EMD. To benchmark cross-dataset metric generalizability under identical conditions, we evaluated our pre-trained backbones on the 54 held-out S&C test videos across 100 stochastic trials. In each trial, $K = 1$ random exemplar video was sampled per class as the support set (yielding 4 support videos total across the 4 S&C classes), while the remaining 50 test videos served as query samples. Classification was performed via nearest-neighbor matching using cosine similarity over the temporal mean-aggregated feature representations. Under this protocol, Transformer Mix achieves a 1-shot video accuracy of **97.32% $\pm$ 1.96%** (95% CI: [94.00%, 98.00%]), AAGCN Bone reaches **92.02% $\pm$ 7.24%**, and SkelGym-Full reaches **95.34% $\pm$ 3.89%** (95% CI: [80.00%, 98.00%]). These results demonstrate that the proposed 117-d compound representation enables robust metric matching even under extreme single-sample constraints.

6 Biomechanical Kinematics and Error Analysis

A primary objective of this work is to bridge computer vision modeling with clinical biomechanics. Below, we conduct an in-depth anatomical analysis of the model’s predictions and confusion patterns (Fig. 5):

6.1 Upper-Limb Kinematic Ambiguity: Biceps Curl vs. Hammer Curl

Inspection of Fig. 5 indicates that the most prominent confusion occurs between *hammer curl* and *barbell biceps curl* (35.4% recall for hammer curl, with 41.2% of its windows misclassified as biceps curl).

Biomechanically, both exercises execute identical sagittal elbow flexion driven by the elbow joint angle $\theta_{\text{elbow}} = \angle(\text{shoulder, elbow, wrist})$ moving through $30^\circ \rightarrow 145^\circ$. The sole physiological differentiator resides in the forearm’s rotational orientation across the radioulnar joint [41,42]. In the barbell biceps curl, the forearm is held in full supination (palms oriented anteriorly and superiorly), which maximally tensions the distal tendon insertion across the radial tuberosity and optimizes mechanical leverage for both the long and short heads of the *biceps brachii*. In contrast, the hammer curl maintains a neutral semi-pronated posture (thumbs directed superiorly), shifting primary torque production to the deeper *brachialis* and the superficial *brachioradialis* muscles along the lateral forearm [41]. Because monocular pose estimators (such as MediaPipe) represent the wrist as a single Euclidean point centroid (x, y, z) without recovering full hand rotational mesh orientation or palm surface normals, the 3D trajectory of the wrist joint center in Cartesian space remains virtually identical between supinated and neutral curls. Resolving this subtle anatomical boundary will require integrating fine-grained hand pose keypoints or wrist rotational orientation sensors.

6.2 Posterior Chain Kinematics: Conventional Deadlift vs. Romanian Deadlift

Another noticeable confusion occurs between *deadlift* and *Romanian deadlift* (RDL). The conventional deadlift represents a ground-based compound movement characterized by substantial knee flexion ($\theta_{\text{knee}} \approx 70^\circ - 90^\circ$ at lift initiation) that recruits the quadriceps femoris to generate initial vertical propulsion, transitioning into hip extension as the bar passes the patellae [37]. In contrast, the Romanian deadlift functions as a pure posterior-chain hip hinge movement initiating from an upright standing lockout [38,36]. Throughout the descent, the knees remain relatively static with slight soft flexion ($\theta_{\text{knee}} \approx 15^\circ - 25^\circ$), with kinematics dominated by maximal anterior pelvic tilt and hip flexion that places intense eccentric loading onto the hamstrings and gluteus maximus. When athletes execute deadlifts with sub-optimal form (e.g., stiff-legged deadlifting with inadequate knee bend) or perform RDLs with excessive knee travel, their sagittal kinematic trajectories overlap significantly. Our 117-d mix feature captures the hip-to-knee elevation angle ($\theta_{\text{hip, knee}}$), enabling the ensemble to separate them correctly in 80% of video consensus cases (Table 10).

6.3 Pressing Plane Angle Analysis: Flat vs. Incline vs. Decline Bench Press

The bench press family targets the *pectoralis major* muscle, with the three variants differing primarily in the inclination angle of the torso plane relative to the gravity vector [39,40]. During the flat bench press ($\alpha \approx 0^\circ$), the horizontal pressing trajectory predominantly engages the sternocostal head of the pectoralis major via horizontal glenohumeral adduction. Elevating the bench inclination ($\alpha \approx +30^\circ$ to $+45^\circ$) shifts prime mechanical recruitment superiorly toward the clavicular head of the pectoralis major and anterior deltoids, causing the humerus to elevate along an oblique superior plane relative to the sternum [40]. Conversely, the decline configuration ($\alpha \approx -15^\circ$) isolates the inferior sternocostal fibers with reduced anterior deltoid contribution [39]. Because relative 3D displacement vectors $\tilde{p}$ are referenced to the nose anchor, the elevation angles $\theta_{\text{shoulder, wrist}}$ and $\theta_{\text{shoulder, hip}}$ accurately reflect bench tilt. As shown in Table 10, the ensemble achieves 64.29% video recall on flat bench, 62.50% on decline bench, and 0.6154 F1 on incline bench, with errors largely confined to camera angles where bench tilt is foreshortened.

6.4 Validation-Grounded Decision Protocol

We emphasize that all model selection choices—including adopting the 117-d mix feature, retaining SkelGym-Aug, and solving SLSQP voting weights—were established strictly on the validation set, where the SkelGym-Full ensemble reached 80.10% window accuracy and 83.09% video accuracy. The subsequent 77.68% test result represents a reliable held-out confirmation of generalization to unseen subjects and recording environments. Finally, out of 236 held-out test videos, exactly 233 contained valid action segments exceeding 32 frames after boundary trimming, explaining the test support count.

7 Discussion, Limitations, and Future Work

Trade-offs Between Kinematic Skeleton Fusion and Dense Video Vision Transformers. While SkelGym deliberately centers on lightweight, coordinate-based 3D skeleton sequences, an alternative foundational paradigm in computer vision relies on dense spatio-temporal video architectures, including 3D CNNs (e.g., SlowFast [3]) and Video Vision Transformers (e.g., Video Swin Transformer / Swin3D [6]). Dense video transformers possess the distinct theoretical advantage of directly ingesting raw appearance pixels, allowing them to exploit environmental context, gym machine geometries, barbell grip orientations, and subtle muscular contractions without relying on intermediate landmark extraction pipelines. This appearance awareness is particularly beneficial for resolving fine-grained ambiguities such as the pronation vs. supination distinction between hammer curls and barbell curls.

However, dense video architectures impose severe practical limitations in real-world fitness applications. First, their computational complexity is substantial:

Video Swin models frequently require tens to hundreds of GFLOPs per snippet, incurring latencies of several hundred milliseconds per prediction window and rendering them completely unsuited for real-time inference on edge smartphones or micro-controllers. In contrast, SkelGym executes in ≈ 1.01 ms per window. Second, RGB video models exhibit severe vulnerability to environmental background and athletic apparel memorization, leading to dramatic generalization collapse when evaluated in unseen home gym settings. Third, cloud-based video analytics require continuously transmitting unconstrained, high-resolution RGB living-room video streams across public networks, introducing severe security vulnerabilities and privacy intrusion concerns for consumer smart-camera devices. In contrast, while monocular pose estimation (such as MediaPipe) still operates on optical camera frames at the physical sensor level, the ultra-low latency of compact skeleton classifiers enables the entire perception pipeline to execute locally on edge hardware. Under this edge-native paradigm, raw RGB frames are processed ephemerally in volatile RAM and immediately discarded frame-by-frame without persistent disk caching or remote network streaming. Only non-identifying, low-dimensional coordinate trajectories (x, y, z) are retained, enabling a privacy-aware workflow that reduces raw RGB retention and prevents continuous cloud video streaming. Consequently, we identify hybrid hierarchical architectures—wherein lightweight skeleton models handle primary real-time classification, triggering localized, low-frequency 3D video attention only when kinematic ambiguity surpasses uncertainty thresholds—as a promising direction for future research.

Limitations of Monocular Pose Estimation and Failure Modes. While monocular pose extraction via MediaPipe Pose provides high throughput on edge processors, it incurs physical and optical failure modes that bound skeleton-based recognition accuracy. First, resistance training environments frequently introduce severe physical occlusions: barbell weight plates (e.g., standard 450 mm Olympic plates) frequently obstruct wrist, elbow, and torso tracking during deadlifts, rows, and bench presses, while spotters standing adjacent to equipment introduce visual occlusion. Second, camera field-of-view boundaries cause transient out-of-frame limb clipping, such as feet moving below frame during Romanian deadlifts or hands ascending above frame during pull-ups and overhead presses. Third, monocular 2D-to-3D landmark lifting estimates metric depth (z) relative to the hip midpoint using geometric heuristics and anthropometric priors. Consequently, movements aligned along the camera’s optical axis exhibit higher longitudinal depth jitter than lateral sagittal plane perspectives. Future iterations could incorporate multi-view spatial consensus or temporal smoothing filters (e.g., learned kinematic Kalman filters) to improve landmark stability.

Dataset Class Imbalance and Long-Tail Kinetics. A second practical limitation concerns the natural class distribution within crowdsourced fitness video datasets. Movement availability reflects gym training popularity: high-frequency compound and aesthetic movements (such as barbell biceps curls and bench presses, each with 84 videos) possess extensive diversity, whereas specialized

or bodyweight stabilization exercises (such as plank with 9 videos) reside in the long tail. Although we mitigated class imbalance during optimization using Macro F1 monitoring, validation-calibrated loss weighting, and label smoothing ($\alpha = 0.05$), rare classes exhibit wider confidence intervals. Expanding the public corpus through standardized multi-angle video collection across underrepresented exercises remains an essential objective for subsequent dataset releases.

Conclusion. In this work, we presented **SkelGym**, a lightweight, computationally efficient, and biomechanically grounded framework for 22-class gym exercise classification. By coupling the proposed 117-dimensional Biomechanical Compound Representation with dynamic SkelGym-Aug and integrating self-attention sequence modeling with 4-stream adaptive graph convolutions via validation-calibrated soft voting, SkelGym attains **70.11% window-level accuracy** and **77.68% video-level consensus accuracy** across 22 resistance exercises. With an ultra-low classifier inference latency of **1.01 ms per 32-frame window** on edge hardware (excluding monocular pose extraction), SkelGym offers a solid foundation for real-time virtual training applications.

Data, Reproducibility, and Code Availability

To facilitate rigorous scientific reproducibility and edge deployment, all project software, extracted kinematic representations, and pre-trained models are publicly hosted on Hugging Face Hub at <https://huggingface.co/Cuong2004/gym-exercise-classification>. The repository includes PyTorch source code for all sequence and spatial-temporal graph backbones (Transformer Mix, ST-GCN, and Four-Stream AAGCN), training and evaluation scripts, and trained model checkpoints.

The complete skeletal dataset is released in JSON, NPY, and HDF5 tensor formats, accompanied by exact temporal action segment boundaries (`.txt`) and consolidated metadata records (`Final_dataset_metadata.csv`) across Hugging Face and Kaggle. Distributing pre-extracted 3D landmark coordinates directly eliminates reliance on third-party video URLs that suffer from link-rot, while ensuring deterministic reproducibility without demanding intensive GPU pose estimation re-runs. In strict compliance with online platform Terms of Service and subject privacy considerations, raw third-party internet video recordings are withheld from redistribution. However, the 128 author-recorded gymnasium video clips (collected with informed consent under institutional ethical oversight) and all 1,024 extracted coordinate sequences are freely accessible for academic and non-commercial research purposes.

References

1. Donahue, J., Hendricks, L.A., Guadarrama, S., Rohrbach, M., Venugopalan, S., Saenko, K., Darrell, T.: Long-term recurrent convolutional networks for visual recognition and description. In: CVPR, pp. 2625–2634 (2015)

2. Carreira, J., Zisserman, A.: Quo vadis, action recognition? A new model and the kinetics dataset. In: CVPR, pp. 6299–6308 (2017)
3. Feichtenhofer, C., Fan, H., Malik, J., He, K.: SlowFast networks for video recognition. In: ICCV, pp. 6202–6211 (2019)
4. Arnab, A., Dehghani, M., Heigold, G., Sun, C., Lučić, M., Schmid, C.: ViViT: A video vision transformer. In: ICCV, pp. 6836–6846 (2021)
5. Bertasius, G., Wang, H., Torresani, L.: Is space-time attention all you need for video understanding? In: ICML, pp. 813–824 (2021)
6. Liu, Z., Ning, J., Cao, Y., Wei, Y., Zhang, Z., Lin, S., Hu, H.: Video Swin Transformer. In: CVPR, pp. 3202–3211 (2022)
7. Ng, J.Y.H., Hausknecht, M., Vijayanarasimhan, S., Vinyals, O., Monga, R., Toderici, G.: Beyond short snippets: Deep networks for video classification. In: CVPR, pp. 4694–4702 (2015)
8. Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.L., Yong, M.G., Lee, J., et al.: MediaPipe: A framework for building perception pipelines. arXiv preprint arXiv:1906.08172 (2019)
9. Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., Grundmann, M.: BlazePose: On-device real-time body pose tracking. In: CVPR Workshop on Computer Vision for Augmented and Virtual Reality (2020)
10. Cao, Z., Hidalgo, G., Simon, T., Wei, S.E., Sheikh, Y.: OpenPose: Realtime multi-person 2D pose estimation using Part Affinity Fields. IEEE TPAMI 43(1), 172–186 (2019)
11. Fang, H.S., Li, J., Tang, H., Xu, C., Zhu, H., Lu, C., Fang, H.S.: AlphaPose: Whole-body knowledge-transfer for online pose estimation. IEEE TPAMI 45(6), 7166–7182 (2022)
12. Hochreiter, S., Schmidhuber, J.: Long short-term memory. Neural Computation 9(8), 1735–1780 (1997)
13. Graves, A., Fernández, S., Schmidhuber, J.: Bidirectional LSTM networks for improved phoneme classification and recognition. In: ICANN, pp. 799–804 (2005)
14. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: NeurIPS, pp. 5998–6008 (2017)
15. Li, C., Wang, P., Wang, S., Hou, Y., Li, W.: Skeleton-based action recognition using LSTM and CNN. In: ICMEW, pp. 585–590 (2017)
16. Yan, S., Xiong, Y., Lin, D.: Spatial temporal graph convolutional networks for skeleton-based action recognition. In: AAAI, vol. 32, pp. 7444–7452 (2018)
17. Shi, L., Zhang, Y., Cheng, J., Lu, H.: Two-stream adaptive graph convolutional networks for skeleton-based action recognition. In: CVPR, pp. 12026–12035 (2019)
18. Shi, L., Zhang, Y., Cheng, J., Lu, H.: Skeleton-based action recognition with multi-stream adaptive graph convolutional networks. IEEE TIP 29, 9532–9545 (2020)
19. Plizzari, C., Cannici, M., Matteucci, M.: Skeleton-based action recognition via spatial and temporal transformer networks. Computer Vision and Image Understanding 208, 103219 (2021)
20. Chen, Y., Zhang, Z., Yuan, C., Li, B., Deng, Y., Hu, W.: Channel-wise topology refinement graph convolution for skeleton-based action recognition. In: ICCV, pp. 13359–13368 (2021)
21. Song, Y.F., Zhang, Z., Shan, C., Wang, L.: Constructing stronger and faster baselines for skeleton-based action recognition. IEEE TPAMI 45(2), 1474–1488 (2023)

22. Ko, H., Lee, J., Kim, Y., et al.: Data augmentation for efficient skeleton-based action recognition: Spatial and temporal approaches. *Electronics* 13(4), 747 (2024)
23. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: *ICLR* (2019)
24. Hendrycks, D., Gimpel, K.: Gaussian error linear units (GELUs). *arXiv preprint arXiv:1606.08415* (2016)
25. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the inception architecture for computer vision. In: *CVPR*, pp. 2818–2826 (2016)
26. Kraft, D.: A software package for sequential quadratic programming. Tech. Rep. DFVLR-FB 88-28, DLR German Aerospace Center (1988)
27. Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., et al.: SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17(3), 261–272 (2020)
28. Wolpert, D.H.: Stacked generalization. *Neural Networks* 5(2), 241–259 (1992)
29. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al.: PyTorch: An imperative style, high-performance deep learning library. In: *NeurIPS*, pp. 8024–8035 (2019)
30. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al.: Scikit-learn: Machine learning in Python. *JMLR* 12, 2825–2830 (2011)
31. Riccio, R.: Real-time fitness exercise classification and counting from video frames. *arXiv preprint arXiv:2411.11548* (2024)
32. Abdillahi, H.: WorkoutFitness Video dataset. *Kaggle* (2023)
33. Ding, H., Liu, C., Wang, H.: RepCount: A large-scale dataset for repetition counting in diverse physical exercises. In: *ACM MM*, pp. 5832–5840 (2022)
34. Soomro, K., Zamir, A.R., Shah, M.: UCF101: A dataset of 101 human actions classes from videos in the wild. *arXiv preprint arXiv:1212.0402* (2012)
35. Kay, W., Carreira, J., Simonyan, K., Zhang, B., Hillier, C., Rao, S., Shukla, P., Heyward, A., Wang, H., Zisserman, A.: The Kinetics human action video dataset. *arXiv preprint arXiv:1705.06950* (2017)
36. Schoenfeld, B.J.: The mechanisms of muscle hypertrophy and their application to resistance training. *Journal of Strength and Conditioning Research* 24(10), 2857–2872 (2010)
37. Escamilla, R.F., Francisco, A.C., Kayes, A.V., Speer, K.P., Moorman, C.T.: A three-dimensional biomechanical analysis of sumo and conventional style deadlifts. *Medicine & Science in Sports & Exercise* 32(7), 1265–1275 (2000)
38. McAllister, M.J., Hammond, K.G., Schilling, B.K., Ferrando, A.A., Webb, H.E.: Muscle activation during various hamstring exercises. *Journal of Strength and Conditioning Research* 28(6), 1573–1580 (2014)
39. Barnett, C., Kippers, V., Turner, P.: Effects of variations of the bench press exercise on the EMG activity of five shoulder muscles. *Journal of Strength and Conditioning Research* 9(4), 222–227 (1995)
40. Trebs, A.A., Brandenburg, J.P., Pitney, W.A.: An electromyography analysis of 3 different incline positions within the dumbbell chest press. *Journal of Strength and Conditioning Research* 24(7), 1925–1930 (2010)
41. Oliveira, L.F., Carregaro, R.L.: Electromyographic activity of biceps brachii and brachioradialis during curls with different hand positions. *Journal of Sports Sciences* 27(13), 1435–1442 (2009)
42. Stienen, P.C., Schless, S.H., Thomason, P., Sangeux, M.: Arm kinematics and forearm rotation in resistance movements. *Sports Biomechanics* 18(4), 412–426 (2019)

43. Deyzel, P., du Preez, J., Engelbrecht, H.: One-shot skeleton-based action recognition on strength and conditioning exercises. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 5313–5322 (2023)

A Anatomical Landmark Mapping and Physiological Significance

Table 13 outlines the 13 anatomical keypoints selected from Google MediaPipe Pose Heavy [8,9]. These joints constitute the core kinematic chain required for analyzing human resistance exercises. Facial landmarks (1–10) and foot extremity points (29–32) are intentionally discarded to eliminate high-frequency tracking noise.

Table 13: Selected 13 anatomical joints from MediaPipe Pose and their kinematic roles.

Joint ID (v)	MediaPipe Index	Anatomical Description	Kinematic Significance in Resistance Training
0	0	Nose	Coordinate origin reference anchor; head orientation tracking
1	11	Left Shoulder	Glenohumeral joint; upper-body pressing/pulling rotation center
2	12	Right Shoulder	Bilateral counterpart to left shoulder
3	13	Left Elbow	Humeroradial joint; elbow flexion/extension in curls/presses
4	14	Right Elbow	Bilateral counterpart to left elbow
5	15	Left Wrist	Radiocarpal joint; barbell/dumbbell contact point trajectory
6	16	Right Wrist	Bilateral counterpart to left wrist
7	23	Left Hip	Acetabulofemoral joint; pelvic hinge pivot in deadlifts/squats
8	24	Right Hip	Bilateral counterpart to left hip
9	25	Left Knee	Tibiofemoral joint; knee flexion/extension in squats/lunges
10	26	Right Knee	Bilateral counterpart to left knee
11	27	Left Ankle	Talocrural joint; ground contact anchor; ankle dorsiflexion
12	28	Right Ankle	Bilateral counterpart to left ankle

B Open Science Model Hub and Checkpoint Loading

All pre-trained PyTorch [29] checkpoints, scripts, and evaluation pipelines are openly accessible on Hugging Face at <https://huggingface.co/Cuong2004/gym-exercise-classification>. Checkpoints can be downloaded in Python:

```
import torch
from huggingface_hub import hf_hub_download

# Download best Transformer Mix checkpoint
ckpt_path = hf_hub_download(
    repo_id="Cuong2004/gym-exercise-classification",
    filename="best_transformer_mix_aug.pt"
)
model = torch.load(ckpt_path, map_location="cpu")
print("Model loaded successfully!")
```